HPSC Food Safety Officer Master Handout — Unit 3, Part B

1. Source Audit & Study Strategy

UNIT 3, PART B: MASTER STUDY HANDOUT

Food Lipids, Deep Frying, Enzyme Kinetics, Natural Pigments, Colours and Fortification

Examination orientation: Haryana Public Service Commission Food Safety Officer, Group B.

Source-audited edition: The fortification provisions in this handout refer to the uploaded compendium, Version III, dated 30 September 2021. They are not represented as a verification of legislation presently in force.

1. What was actually available

Source	Availability	Principal use
Food Chemistry, Satish Parmar, Amit Kumar Jain and K. D. Aparnathi	Uploaded and readable	Lipid classification, frying chemistry, enzymes, flavour enhancers, sweeteners and food colours
Food Safety and Standards (Fortification of Foods) Regulations, 2018, compendium dated 30 September 2021	Uploaded and readable	Definitions, general principles, quality assurance, packaging, labelling and Schedules I–III
Food Science, B. Srilakshmi	Not uploaded in this conversation	No claims, quotations or page references are attributed to this book

Evidence labels used throughout

- **Uploaded-source finding:** Directly supported by one of the two available documents.
- **Chemical clarification:** A first-principles explanation, mathematical derivation or correction of an oversimplified textbook statement. It is not presented as a quotation from an unavailable textbook.
- **Source gap:** A requested detail that the available documents do not establish. Such a detail must not be memorised as verified statutory wording.

The requested scope exceeds the supplied corpus in several places. Those boundaries are identified rather than concealed. In particular, the uploaded fortification compendium is not the statutory instrument establishing frying-oil limits or general food-colour limits.

2. Source navigation

Topic	Uploaded textbook location
Lipid classification, composition and unsaponifiable matter	Lesson 8, printed pages 48–52
Reactions during deep frying	Lesson 9, printed pages 53–55
Lipoproteins and membrane organisation	Lesson 10, printed pages 56–58

Topic	Uploaded textbook location
Proteolysis and chymosin	Lesson 6, printed pages 43–44; Lesson 15, pages 86–87
Enzyme architecture and classification	Lesson 15, printed pages 82–85
Industrial enzymes and process indicators	Lesson 15, printed pages 85–88; Lesson 16, page 91
Substrate concentration, inhibitors, temperature and hydrogen ion concentration	Lesson 16, printed pages 89–93
Nutrient addition	Lesson 17, printed pages 94–98; use the compendium for binding fortification standards
Flavour and flavour enhancers	Lesson 18, printed pages 99–102
Sugar substitutes	Lesson 19, printed pages 103–105
Food colours	Lesson 20, printed pages 106–110
Pigment changes during ripening and tea processing	Lesson 26, pages 139–140; Lesson 29A, page 157

3. High-priority correction register

[HPSC EXAM TRAP] Source authority outranks a coaching-style summary. A confidently worded claim is not statutory proof. Where a textbook and the uploaded regulation differ, use the regulation for the legal answer.

Requested or textbook formulation	Exam-safe correction
The fortification tagline is mandatory	Regulation 7(2) says the package **may also carry** the tagline. The fortification declaration and prescribed logo are required, subject to the iodized-salt exception
Every fortified food must carry the fortification declaration and logo	Iodized salt fortified with iodine is expressly exempted from Regulation 7(2)
High Fat Sugar Salt foods are excluded from every possible meaning of fortified food	Regulation 3(7) excludes them from the **Fortified Processed Foods category**
Manufacture-level iodine must always be 20–30 parts per million	The uploaded August 2021 amendment substitutes 15–30 parts per million on a dry-weight basis
All permitted iron sources for atta are also permitted for rice	The rice table lists ferric pyrophosphate or sodium feredetate; the atta and maida tables list additional iron compounds
All inorganic food additives are prohibited except titanium dioxide in chewing gum	A colour prohibition is not a prohibition on every inorganic additive. Schedule I separately permits titanium dioxide in double fortified salt
The logo dimensions and Pantone 3005 C specification can be quoted from the supplied text	The accessible Schedule II extraction does not establish those numerical specifications; do not invent them
Total Polar Compounds limits of 15% and 25% are established by these uploads	Neither uploaded document establishes those limits. They remain requested external regulatory targets requiring their governing instrument
A general colour limit of 100 milligrams per kilogram, with 200 in specified canned products, is established by these uploads	These general limits and their precise category exceptions are not established by the supplied documents
The Michaelis constant is always an equilibrium dissociation constant	In the usual steady-state model, it is a kinetic constant. Equality with a binding dissociation constant requires a

Requested or textbook formulation	Exam-safe correction
	limiting assumption
Noncompetitive inhibition means any inhibitor binding away from the active site	The unchanged Michaelis constant applies to **pure** noncompetitive inhibition; unequal binding to free enzyme and enzyme–substrate complex gives mixed inhibition
Betalains are chemically indigoid pigments	The textbook uses this terminology, but modern chemical classification identifies them as nitrogen-containing pigments derived from tyrosine and betalamic acid
All carotenoid-related pigments have exactly forty carbon atoms	The forty-carbon structure describes the parent carotenoid framework; cleavage products such as bixin and norbixin are exceptions
All intense sweeteners are excreted unchanged	Aspartame is metabolised. Intense sweetness does not imply identical metabolism
Peroxide and Polenske values can both be abbreviated without clarification	Write their names in full; the shared abbreviation creates an avoidable ambiguity

4. Examination architecture

The following pattern is the one supplied in the request; it has not been independently checked against a recruitment notification.

Stage	Supplied pattern	Preparation implication
Objective screening	100 marks; penalty of 0.25 mark for a wrong answer	Learn definitions, units, source-specific exceptions and pair-counting
Subject Knowledge Test	150 marks; three hours	Explain mechanisms, draw compact pathways, compare analytical measures and quote statutory units accurately
Final merit	Subject Knowledge Test described as carrying 87.5% weightage	Give descriptive practice substantial attention; do not neglect screening qualification

Time budgeting for descriptive writing

Three hours equals 180 minutes. Dividing by 150 marks gives an average of **1.2 minutes per mark**, before allowing for reading and review.

- A ten-mark answer has an approximate gross allocation of twelve minutes.
- A fifteen-mark answer has an approximate gross allocation of eighteen minutes.
- Reserve some total time for question selection and final checking.

A topper-style learning sequence

1. Identify the molecular structure or food vehicle.
2. Explain the reaction or nutritional purpose.
3. Connect the mechanism to an observable processing effect.
4. Select the appropriate analytical measurement.
5. Add the applicable legal provision, with its unit and exception.
6. Close with a qualified conclusion rather than an absolute statement.

Objective-test elimination discipline

For a question asking how many pairs are correct:

1. Judge every pair independently.
2. Mark correct pairs as 1 and incorrect pairs as 0.
3. Add the marks only after completing all judgments.
4. Recheck the words **only**, **always**, **all**, **unchanged**, **mandatory** and **per kilogram**.

If a question awards one mark for a correct answer and deducts 0.25 for an incorrect answer, the expected score from answering is:

$$E = p - 0.25(1-p) = 1.25p - 0.25$$

It becomes positive when the probability of correctness exceeds 0.20. This is an arithmetic observation, not a guarantee that indiscriminate guessing improves a particular examination result. Follow the actual examination instructions.

5. Cross-module conceptual bridge

```
Lipid structure
  |
  +--> Unsaturation --> Oxidation susceptibility --> Peroxide formation
  |                                  |
  |                                  +--> Off-flavour products
  |
  +--> Ester bonds --> Hydrolysis --> Free fatty acids --> Lower smoke point
  |
  +--> Radical reactions at high temperature --> Polymerisation --> Viscosity

Enzyme structure --> Catalytic activity --> Processing benefit or deterioration

Pigment structure --> Response to acid, oxygen and heat --> Visible colour change

Fortificant chemistry + food vehicle + processing + packaging
  |
  +--> Nutrient retention, uniformity, sensory compatibility and compliance
```

6. Rules for using this handout

- Scientific symbols are defined when introduced; prose uses full names wherever practical.
- Approximate textbook values are not treated as universal compositional laws.
- Educational numerical examples are explicitly identified as hypothetical.
- No unsupported statutory threshold is used as the foundation of a scored practice question.
- The 2021 compendium must be distinguished from any later amendment or current enforcement interpretation.

2. Module 5 — Lipids & Analysis

MODULE 5: FOOD LIPIDS, AUTOXIDATION AND ANALYTICAL CONSTANTS

5.1. First principle: why lipids behave differently from sugars

A sugar molecule presents many groups that interact favourably with water. A triacylglycerol presents three long hydrocarbon chains and relatively few polar groups. Water therefore does not dissolve bulk fat readily.

Uploaded-source finding: The textbook classifies lipids primarily through their hydrophobic character and solubility in organic solvents, rather than through one universal structural feature. Some lipids are amphiphilic: they contain both water-compatible and water-incompatible regions.

Functions in food and nutrition

- Supply approximately nine kilocalories per gram.
- Supply essential fatty acids and carry fat-soluble vitamins.
- Contribute body, tenderness, texture and mouthfeel.
- Carry or generate characteristic flavour compounds.
- Contribute to satiety.
- Participate in emulsions and biological membranes when amphiphilic.

Source: Lesson 8, pages 48–49.

5.2. Structural taxonomy

Intuitive opening

Classify a lipid by asking what appears when its major linkages are broken: only fatty acid and alcohol components, additional chemical groups, or a lipid-like constituent that is not itself a simple or compound lipid.

Class	Structural principle	Examples	Examination distinction
Simple lipids	Fatty acids associated with alcohol components, commonly through ester bonds	Triacylglycerols, oils, fats and wax esters	A wax is not a triacylglycerol
Compound lipids	Additional groups accompany the lipid framework	Glycerophospholipids, glyceroglycolipids and sphingolipids	The additional group may be phosphate or carbohydrate
Derived lipids	Lipid constituents or related molecules outside the preceding two groups	Fatty acids, sterols, alcohols, fat-soluble vitamins, pigments and hydrocarbons	The term does not mean that every member must arise through laboratory hydrolysis

Triacylglycerol architecture

A glycerol molecule has three hydroxyl groups. Each can form an ester bond with a fatty acid.

```
Glycerol hydroxyl groups + three fatty acid carboxyl groups
                  |
                  | Esterification
                  v
      Triacylglycerol + three water molecules
```

The three fatty acid chains may be identical or different. Their chain lengths, double bonds and positions on glycerol influence melting behaviour and processing performance.

Compound lipids: structural interpretation

Lipid family	Main framework	Characteristic feature	Functional consequence
Glycerophospholipids	Glycerol, fatty acid residues and a phosphate-containing head group	Polar head and nonpolar tails	Membrane formation and interfacial activity
Glyceroglycolipids	Glycerol-based lipid with carbohydrate groups	Sugar-containing head region	Important in plant membranes
Sphingolipids	Sphingoid-base framework	Not necessarily based on glycerol	Includes sphingomyelin and glycosphingolipids

Chemical clarification: Sphingolipid, phospholipid and glycolipid labels can overlap. Sphingomyelin is a sphingolipid containing phosphate; a glycosphingolipid is both a sphingolipid and a glycolipid. Treating these as universally exclusive boxes is incorrect.

Polarity and saponification are different classification axes

Question	Classification
Does the molecule contain a substantial polar region?	Polar or nonpolar lipid
Does alkaline treatment produce fatty acid salts from its relevant components?	Saponifiable or unsaponifiable fraction

Chemical clarification: Polar does not necessarily mean a molecule carries a net electrical charge. Zwitterionic phospholipids have charged groups but may have zero net charge.

[HPSC EXAM TRAP] Cholesterol is unsaponifiable, but the fatty acid ester bond of a cholesterol ester can be hydrolysed. The released cholesterol subsequently enters the unsaponifiable fraction.

5.3. Fatty acid profiling

Intuitive opening

Think of a fatty acid as a carbon chain ending in a carboxyl group. Chain length affects volatility and melting; double bonds affect packing and oxidation.

The notation **C18:2** means eighteen carbon atoms and two carbon–carbon double bonds. It does not, by itself, identify their positions or geometry.

Fatty acid	Carbon notation	Unsaturation class	Omega family	High-yield significance
Butyric acid	C4:0	Saturated	Not applicable	Characteristic short-chain constituent of milk fat
Caproic acid	C6:0	Saturated	Not applicable	Short-chain milk-fat constituent
Caprylic acid	C8:0	Saturated	Not applicable	Relevant to volatile fatty acid analysis
Capric acid	C10:0	Saturated	Not applicable	Present in milk fat and lauric fats

Fatty acid	Carbon notation	Unsaturation class	Omega family	High-yield significance
Lauric acid	C12:0	Saturated	Not applicable	Characteristic of coconut and palm-kernel fats
Palmitic acid	C16:0	Saturated	Not applicable	Major milk-fat and animal-fat constituent
Stearic acid	C18:0	Saturated	Not applicable	Common long-chain saturated fatty acid
Oleic acid	C18:1	Monounsaturated	Omega-9	Major unsaturated constituent of many edible fats
Linoleic acid	C18:2	Polyunsaturated	Omega-6	Essential fatty acid
Alpha-linolenic acid	C18:3	Polyunsaturated	Omega-3	Essential fatty acid; oxidation-related flavour reversion is important in soybean oil
Arachidonic acid	C20:4	Polyunsaturated	Omega-6	Longer-chain member of the omega-6 family

Evidence boundary: The uploaded textbook establishes the broad fatty acid groups and milk-fat profile. The complete omega-position table above is chemical nomenclature clarification, not a reproduced table from the upload.

Omega notation

Count from the terminal methyl end to the first double bond. Omega-3 and omega-6 describe the location of that first double bond, not the total number of double bonds.

Why polyunsaturated fatty acids are oxidation-sensitive

Hydrogen abstraction is favoured at positions that form resonance-stabilised radicals. Bis-allylic positions between double bonds are particularly vulnerable. This explains why additional unsaturation generally increases susceptibility to autoxidation under comparable conditions.

A practical comparison must still account for antioxidants, oxygen exposure, temperature, metals and the food matrix.

[HPSC EXAM TRAP] “Every unsaturated oil oxidises faster than every saturated fat under all conditions” is too absolute. Molecular susceptibility and actual storage stability are not identical.

5.4. Milk-fat uniqueness and traditional fat groups

Intuitive opening

Milk fat contains many long-chain fatty acids, but its analytically useful fingerprint includes appreciable short-chain fatty acids. A diagnostic minor constituent need not be the most abundant constituent.

Uploaded-source findings:

- The principal milk-fat fatty acids include palmitic, stearic and oleic acids.
- Milk fat also contains appreciable butyric, caproic, caprylic and capric acids.
- Butyric acid is characteristic of milk fat.

Traditional group	Textbook characteristic	Examples or significance
Milk fats	Long-chain major constituents plus appreciable short-chain acids	Butter and ghee
Lauric fats	Lauric acid approximately forty to fifty percent of fatty acids in the textbook description	Coconut and palm-kernel-type fats
Vegetable butters	Narrow melting range related to triacylglycerol composition and arrangement	Cocoa butter
Oleic–linoleic fats	Substantial oleic and linoleic acids	Many common seed oils
Linolenic fats	Appreciable linolenic acid	Soybean, rapeseed and wheat-germ oils
Animal depot fats	Considerable saturated fatty acids and saturated triacylglycerols	Lard and tallow
Marine oils	Long-chain omega-3 polyunsaturated fatty acids, potentially with several double bonds	Fish oils

Chemical clarification: Low melting in lauric fats is not explained by low unsaturation. Chain length and triacylglycerol composition must also be considered. Palm oil and palm-kernel oil are chemically different.

Selected composition from the textbook

Lipid constituent	Cow milk fat, percent by mass	Buffalo milk fat, percent by mass	Human milk fat, percent by mass
Triacylglycerols	97.5	98.6	98.2
Diacylglycerols	0.36	0.4	0.7
Monoacylglycerols	0.02	0.03	Traces
Cholesterol	0.31	0.3	0.25
Phospholipids	0.6	0.5	0.26
Free fatty acids	0.027	0.5	0.4

These are textbook compositional figures, not universal specifications for all milk-fat samples.

Source: Lesson 8, pages 49–50.

5.5. Unsaponifiable matter

Intuitive opening

During saponification, esterified fatty acids are converted into soaps. A small collection of lipid-associated substances does not become fatty acid soap and can be recovered as the unsaponifiable fraction.

Uploaded-source finding: Fats and oils contain an average of approximately **0.2–1.5%** unsaponifiable compounds in the textbook account. This is a descriptive range, not a universal statutory limit.

Constituent	Chemical feature	Occurrence or role
Cholesterol	Sterol with four fused rings	Predominant sterol in animal fats
Beta-sitosterol	Phytosterol	Predominant phytosterol in many

Constituent	Chemical feature	Occurrence or role
		vegetable oils
Campesterol and stigmasterol	Phytosterols	Additional plant sterols
Squalene	Thirty-carbon linear triterpene hydrocarbon	Important in olive oil, rice oil and fish-liver oil
Tocopherols and tocotrienols	Redox-active vitamin E-related compounds	Biological activity and antioxidant action
Carotenoids	Conjugated polyene pigments	Yellow, orange and red colours
Aliphatic and terpenic alcohols	Lipid-associated alcohols	Components of the residual fraction

Sterol acetate melting-point test

Acetate	Textbook melting point
Cholesterol acetate	Approximately 114 degrees Celsius
Phytosterol acetates	Approximately 126–137 degrees Celsius

Principle: Convert isolated sterols into their acetates and examine their melting behaviour. A higher-melting phytosterol-containing fraction can indicate vegetable-fat addition to an animal fat.

Limitations: Mixtures, purification, recrystallisation and the fat's natural composition affect interpretation. This is not an infallible standalone identification of every adulterant.

Squalene and vitamin-related distinctions

- Squalene has thirty carbon atoms, not forty.
- The textbook reports squalene at approximately one to seven grams per kilogram in olive oil and approximately 3.3 grams per kilogram in rice oil.
- Cholesterol participates in steroid and bile-acid biosynthesis.
- Irradiation of ergosterol produces ergocalciferol, vitamin D2.
- Photochemical conversion of 7-dehydrocholesterol produces cholecalciferol, vitamin D3.
- The fortification compendium nevertheless specifies **plant-source vitamin D** for the oil and milk vehicles covered by its tables.

Source: Lesson 8, pages 50–52; fortification Schedule I.

5.6. Autoxidation: the free-radical chain reaction

Intuitive opening

A radical is a species containing an unpaired electron. Once a lipid radical reacts with oxygen, the resulting peroxyl radical can remove hydrogen from another lipid molecule. One initiation event can therefore sustain many propagation steps.

Let **RH** represent an oxidisable lipid and **R•** its carbon-centred radical.

```
INITIATION
RH + initiating radical --> R• + hydrogenated initiator

PROPAGATION
R• + O2 --> ROO•
ROO• + RH --> ROOH + R•

TERMINATION
```

```
R• + R• --> nonradical product
R• + ROO• --> nonradical product
ROO• + ROO• --> nonradical products

SECONDARY DECOMPOSITION
ROOH --> reactive intermediates --> aldehydes, ketones,
                                    alcohols, acids and hydrocarbons
```

Stepwise interpretation

1. **Initiation:** Heat, light-related processes or metal-catalysed reactions help generate radicals or decompose pre-existing hydroperoxides.
2. **Propagation:** The lipid radical adds oxygen. The peroxyl radical generates a hydroperoxide and another lipid radical.
3. **Termination:** Two radicals react, removing chain carriers.
4. **Hydroperoxide decomposition:** Primary oxidation products break down into secondary products, many of which produce objectionable odours.

Hydroperoxides are generally not the principal volatile rancid-odour compounds. Their decomposition products are central to rancid flavour.

Important controls

Factor	Mechanistic importance	Control principle
Oxygen	Reactant in propagation	Limit headspace oxygen and oxygen entry
Heat	Accelerates reactions and decomposition	Avoid unnecessary thermal exposure
Light	Can initiate photochemical pathways	Use light-protective packaging where appropriate
Iron and copper	Promote oxidation and hydroperoxide decomposition	Avoid contamination and unsuitable contact surfaces
Antioxidants	Can interrupt radical propagation	Use appropriate, legally permitted antioxidants
Water activity	Alters catalysts, mobility and interfacial behaviour	Do not assume that drying always prevents oxidation

Uploaded-source link: Lesson 2, pages 22–23 describes a nonmonotonic relationship between water activity and lipid oxidation. The approximate minimum near water activity 0.3–0.4 is a teaching pattern, not a universal constant.

Antioxidant mechanism

```
Lipid peroxyl radical + antioxidant hydrogen donor
                  |
                  v
Lipid hydroperoxide + relatively stabilised antioxidant radical
```

Chemical clarification: Photooxidation, especially singlet-oxygen oxidation, is not identical to the ordinary free-radical autoxidation sequence.

[HPSC EXAM TRAP] A low peroxide value can describe fresh oil **or** extensively heated oil whose hydroperoxides have decomposed. Interpret it with processing history and other measurements.

5.7. Hydrolytic and ketonic rancidity

Hydrolytic rancidity

Water breaks ester linkages; lipases accelerate the reaction in suitable conditions.

```
Triacylglycerol + water --> diacylglycerol + free fatty acid
Diacylglycerol + water --> monoacylglycerol + free fatty acid
Monoacylglycerol + water --> glycerol + free fatty acid
```

The complete net hydrolysis consumes three water molecules per triacylglycerol.

Consequences:

- Increased acidity and acid value.
- Objectionable flavour when sufficient short-chain fatty acids are released.
- Reduced smoke point in frying oil.
- Desirable flavour development when carefully controlled during certain cheese-ripening processes.

At deep-frying temperatures, substantial hydrolysis does not require survival of the original food lipase; heat and food-derived water are sufficient contributors.

Ketonic rancidity

Source gap: Ketonic rancidity is not developed in the supplied textbook sections. The following is a chemical clarification of the requested term, not an attributed textbook passage.

Certain fungi can convert fatty acids through incomplete beta-oxidative pathways into beta-keto acids. Decarboxylation produces methyl ketones.

```
Fatty acid --> beta-keto acid --> methyl ketone + carbon dioxide
```

A beta-keto acid can be represented as:

```
R-CO-CH2-COOH --> R-CO-CH3 + CO2
```

Medium-chain substrates are important. Related methyl-ketone formation can be desirable in blue-cheese flavour, while uncontrolled formation elsewhere is a defect.

Feature	Autoxidative rancidity	Hydrolytic rancidity	Ketonic rancidity
Core event	Oxygen-associated oxidation	Ester-bond cleavage	Microbial conversion to methyl ketones
Important intermediates	Radicals and hydroperoxides	Partial glycerides and free fatty acids	Beta-keto acids
Main diagnostic direction	Primary and secondary oxidation measurements	Acid value or free fatty acid measurement	Product-specific methyl-ketone profile
Always undesirable?	Usually undesirable in edible-fat storage	No; controlled lipolysis can develop cheese flavour	No; controlled formation contributes to some cheese flavours

5.8. Analytical lipid constants

Evidence boundary

The uploaded textbook directly defines the saponification value and discusses changes in iodine value, free fatty acids and peroxide content during frying. It does **not** supply a complete laboratory treatment of all five requested constants. The method principles and equations below are analytical chemistry explanations; Reichert–Meissl and Polenske details require the missing prescribed analytical or textbook source for strict source-only attribution.

Intuitive opening

Each analytical constant answers a different question. A high value is not inherently good or bad: its meaning depends on what is measured.

Constant	Question answered	Unit or defined basis
Saponification value	How much alkali is consumed by one gram of fat?	Milligrams of potassium hydroxide per gram
Iodine value	How much halogen-equivalent uptake reflects unsaturation?	Grams of iodine per one hundred grams of fat
Peroxide value	How much primary peroxide-related oxidation is detected?	Milliequivalents of active oxygen per kilogram
Reichert–Meissl value	How much steam-volatile, water-soluble fatty acid is recovered under the prescribed method?	Millilitres of 0.1 normal alkali for the defined five-gram fat sample
Polenske value	How much steam-volatile, water-insoluble fatty acid is recovered under the prescribed method?	Millilitres of 0.1 normal alkali for the defined five-gram fat sample

A. Saponification value

Definition: Milligrams of potassium hydroxide required to saponify one gram of fat under the prescribed conditions.

Principle: Reflux the fat with a known excess of alcoholic potassium hydroxide. Back-titrate unused alkali with standard acid. A reagent blank measures the initial alkali available.

Triacylglycerol + 3 KOH --> glycerol + 3 potassium fatty acid salts

$$Saponification\ value = \frac{56.1(B-S)N}{m}$$

Where:

- **B** is the blank acid titre in millilitres.
- **S** is the sample acid titre in millilitres.
- **N** is the acid normality.
- **m** is sample mass in grams.
- **56.1** is the relevant potassium hydroxide mass-conversion factor.

For an ideal pure triacylglycerol:

$$Saponification\ value \approx \frac{168300}{M}$$

Here **M** is its molar mass in grams per mole. Three ester bonds account for the factor of three.

Interpretation: Shorter average fatty acid chains generally mean more triacylglycerol molecules per gram and therefore a higher saponification value.

Qualification: Free fatty acids, unsaponifiable matter and degradation products affect the relationship. It is not a direct chain-length ruler for every degraded or mixed sample.

Hypothetical worked example: Blank titre 25.0 millilitres; sample titre 10.0 millilitres; acid normality 0.5; fat mass 2.0 grams.

$$Saponification\ value = \frac{56.1(25.0-10.0)(0.5)}{2.0} = 210.375$$

Report approximately **210.4 milligrams of potassium hydroxide per gram**.

B. Iodine value

Definition: Grams of iodine absorbed, expressed as iodine equivalent, by one hundred grams of fat under the prescribed method.

Principle: Halogen reagent reacts with carbon–carbon double bonds. The unreacted reagent is determined iodometrically.

Method	Characteristic reagent
Wijs	Iodine monochloride in acetic acid
Hanus	Iodine monobromide in acetic acid

$$Iodine\ value = \frac{12.69(B-S)N}{m}$$

Here the titres are thiosulphate titres in millilitres, **N** is thiosulphate normality and **m** is sample mass in grams.

Interpretation: Greater double-bond content per unit mass generally gives a higher iodine value. The fatty acid chain length and sample composition also influence the result.

Frying connection: Oxidative and polymerisation reactions consume or alter unsaturation, so iodine value commonly decreases.

[HPSC EXAM TRAP] Iodine value measures a bulk unsaturation-related property; it does not identify the omega family or distinguish every cis and trans isomer.

C. Peroxide value

Definition: A primary oxidation index conventionally expressed as milliequivalents of active oxygen per kilogram of fat.

Principle: Peroxide-related oxidants liberate iodine from iodide under the method conditions. Titrate the liberated iodine with sodium thiosulphate.

```
ROOH + 2 I- + 2 H+ --> ROH + I2 + H2O
I2 + 2 thiosulphate ions --> 2 I- + tetrathionate ion
```

$$Peroxide\ value = \frac{1000(S-B)N}{m}$$

Where sample and blank titres are in millilitres and sample mass is in grams.

Hypothetical worked example: Sample titre 1.50 millilitres; blank titre 0.10 millilitre; thiosulphate normality 0.01; sample mass 5.0 grams.

$$Peroxide\ value = \frac{1000(1.50-0.10)(0.01)}{5.0} = 2.8$$

Result: **2.8 milliequivalents per kilogram**.

Limitation: Formation and destruction occur simultaneously. A decreasing peroxide value during heating does not prove restoration of quality.

D. Reichert–Meissl value

Definition: Millilitres of 0.1 normal alkali needed to neutralise the steam-volatile, water-soluble fatty acids obtained from five grams of fat under the prescribed procedure.

Principle: Saponify, acidify to release fatty acids, distil under specified conditions, separate the appropriate soluble fraction and titrate it.

Principal teaching association: Butyric and caproic acids; useful in examination discussions of butter and ghee identity and purity.

For a complete recovered fraction:

$$Value = (S-B)\frac{N}{0.1}$$

If only an aliquot is titrated, multiply by the prescribed total-volume-to-aliquot factor. In a method using a 100-millilitre aliquot from 110 millilitres, that factor is 1.1. Do not insert this factor into every method indiscriminately.

Qualification: This is an operationally defined distillation value, not an exhaustive measurement of every water-soluble fatty acid in the original fat.

E. Polenske value

Definition: Millilitres of 0.1 normal alkali needed to neutralise the steam-volatile, water-insoluble fatty acids obtained from five grams of fat under the prescribed procedure.

Principle: Recover the insoluble volatile acid fraction, dissolve it in the specified solvent and titrate.

Principal teaching association: Caprylic and capric acids, with contributions depending on the method; useful for recognising lauric-fat addition.

For the complete recovered fraction:

$$Value = (S-B)\frac{N}{0.1}$$

Important distinction: Coconut and **palm-kernel** fats are the relevant lauric-fat comparison. Plain palm oil should not be substituted for palm-kernel oil in a mechanistic explanation.

F. Acid value: the essential companion measurement

Definition: Milligrams of potassium hydroxide required to neutralise the free acids in one gram of sample.

$$Acid\ value = \frac{56.1(S-B)N}{m}$$

Unlike saponification value, acid value does not deliberately count the alkali required to hydrolyse all ester bonds.

If free fatty acids are reported **as oleic acid**:

$$Free\ fatty\ acids\ (percent) = \frac{28.25(S-B)N}{m}$$

Approximately:

$$Free\ fatty\ acids\ as\ oleic\ acid\ (percent) = \frac{Acid\ value}{1.99}$$

The conversion changes when a different reference fatty acid is used.

Final diagnostic comparison

Observation	Best initial interpretation	What it does not prove
High saponification value	Lower average molar mass, subject to composition	Specific adulterant identity
High iodine value	Greater unsaturation-related uptake	High nutritional quality in every respect
High peroxide value	Accumulated primary oxidation products	Complete extent of all secondary oxidation
High acid value	More free acid	Oxidation as the only cause
Characteristic Reichert–Meissl value	Compatible volatile soluble acid profile	Absolute purity without other evidence
Elevated Polenske value in suspected adulteration	Possible altered volatile insoluble acid profile	Coconut as the only possible explanation

Descriptive-answer closing line: Lipid quality is multidimensional; a defensible conclusion combines composition, hydrolysis, primary oxidation, secondary degradation and the sample's processing history.

3. Deep-Frying Dynamics

DEEP-FAT FRYING: TRANSPORT, REACTION AND QUALITY CONTROL

Core source: *Food Chemistry*, Lesson 9, printed pages 53–55.

1. First principle: frying is not merely heating food in oil

A fryer is simultaneously a heat-transfer system, a water-removal system and a chemically reactive lipid environment. Food introduces moisture and solids; the oil supplies heat and becomes part of the finished product.

The textbook describes frying at approximately **180 degrees Celsius**. This is a representative operating temperature, not a legal requirement that every food must be fried at exactly that temperature.

2. Physical events at the food–oil interface

```
Food enters hot oil
    |
    +--> Heat enters the food
    |
    +--> Water moves outward and becomes steam
    |       |
    |       +--> Agitation of oil
    |       +--> Steam stripping of volatile compounds
    |       +--> Water available for hydrolysis
    |
    +--> Crust and pore structure develop
    |
    +--> Oil is carried with or absorbed into the product
    |
    +--> Food lipids and solids may enter the frying medium
```

Steam distillation of volatiles

Water released by food sweeps some volatile oxidation products out of the oil. Their concentration may approach a plateau when formation is balanced by evaporation, stripping or further reaction.

Critical inference: Removal of volatile compounds does not remove the entire nonvolatile degradation burden.

Steam blanket

The textbook notes that steam above the oil can reduce oxygen availability at the surface. This protective tendency does not eliminate oxidation throughout the fryer.

Fat uptake

Uploaded-source finding: Finished fried foods may contain approximately **5–40% fat by mass**; potato chips are illustrated at approximately 35%.

These are broad product examples. They are not a universal absorption fraction for every food, nor are they a legal maximum.

Chemical interpretation: Product porosity, surface condition, drainage and cooling affect retained oil. The amount measured includes more than one possible uptake mechanism.

Lipid exchange

Chicken and other fatty foods can release endogenous lipids into the oil. The resulting frying medium may have a different fatty acid composition and oxidative stability from the starting oil.

3. Three major reaction families

Reaction family	Main drivers	Representative products	Quality consequences
Hydrolysis	Food-derived water and heat	Free fatty acids, monoacylglycerols, diacylglycerols and glycerol	Increased acidity and lower smoke point
Thermal oxidation	High temperature, oxygen and catalytic contaminants	Hydroperoxides, carbonyl compounds, hydroxy compounds, epoxy compounds and cleavage products	Flavour change and accumulation of oxidised products
Polymerisation and thermal alteration	Radical coupling and reactions of unsaturated lipid chains	Dimers, trimers, cyclic monomers and higher polymers	Increased viscosity, gums, deposits and persistent foaming

4. Thermal oxidation

Intuitive opening

At room temperature, hydroperoxides may accumulate long enough to be measured. At frying temperature, they can decompose rapidly. The balance shifts toward secondary and nonvolatile products.

```
Unsaturated lipid + oxygen
        |
        v
Hydroperoxides and related intermediates
        |
        +--> Cleavage --> volatile aldehydes, ketones, acids and alcohols
        |
        +--> Oxygenated nonvolatile compounds
        |
        +--> Radical formation --> coupled and polymerised products
```

Product classes identified by the textbook

1. Volatile compounds: saturated and unsaturated aldehydes, ketones, hydrocarbons, lactones, alcohols, acids and esters.
2. Nonpolymeric polar compounds: examples include hydroxyl and epoxy acids.
3. Dimeric and polymeric acids and glycerides.
4. Free fatty acids from hydrolysis and other degradation routes.

Peroxide-value paradox

The textbook explicitly describes peroxide content increasing after an induction period and subsequently decreasing.

[HPSC EXAM TRAP] “The peroxide value of repeatedly heated oil must increase continuously” is incorrect. Hydroperoxide decomposition can outpace formation.

5. Hydrolysis and smoke point

Food continually supplies water. Cleavage of ester bonds releases free fatty acids and partial glycerides.

Uploaded-source finding: Liberation of free fatty acids causes a decrease in smoke point.

Chemical clarification: Smoke point also depends on the oil's refining history and other components. A free-fatty-acid concentration is not the sole universal determinant.

```
More water exposure + thermal history
            |
            v
More ester cleavage and free fatty acids
            |
            +--> Greater acidity
            +--> Greater tendency to smoke at lower temperature
```

6. Polymerisation

Intuitive opening

When reactive lipid chains join, the average molecular size rises. Large molecules move less easily, so viscosity increases.

The textbook describes cyclic and noncyclic products involving carbon–carbon linkages and oxygen bonding.

Product type	Meaning
Cyclic monomer	One altered lipid-derived unit containing newly formed cyclic structure
Dimer	Two units coupled together
Trimer	Three units coupled together
Higher polymer	Larger network or chain of coupled units

Polymerisation tends to increase with unsaturation and frying time under the textbook conditions.

Consequences:

- Increased molecular size and viscosity.
- Foaming and gum accumulation.
- Darker oil.
- Altered heat transfer.
- Increased oil retention by food in the textbook description.
- Greasy, less acceptable finished products when deterioration is excessive.

7. Observable and analytical changes

Parameter	Typical textbook direction during substantial deterioration	Qualification
Viscosity	Increases	Strongly related to polymerised products
Foaming tendency	Increases	Persistent chemical foaming must be distinguished from normal steam bubbling
Colour darkness	Increases	Food particles and food-derived compounds also contribute

Parameter	Typical textbook direction during substantial deterioration	Qualification
Free fatty acids	Increase	Extent depends on water, temperature and turnover
Smoke point	Decreases	Not a standalone legal discard test
Iodine value	Decreases	Consistent with altered or consumed unsaturation
Refractive index	Often increases in the textbook account	Temperature control is needed in measurement
Peroxide content	Can rise and then fall	Unsuitable as the sole assessment of very hot oil
Saponification value	Increase reported for accumulated degradation products	Do not infer that every polymerisation event must increase it

Reconciling polymerisation with saponification-value changes

Polymerisation alone increases molecular size and can suggest a lower alkali demand per unit mass. Frying, however, produces a mixture that also includes cleavage products and free acids. The net measured direction reflects competing processes. This is why the ideal pure-triacylglycerol equation must not be applied mechanically to degraded frying oil.

8. Continuous versus intermittent frying

Feature	Continuous frying	Intermittent frying
Oil replacement	Often substantial because product removes oil	May be slower or irregular
Heating–cooling cycles	Fewer within a given production run	Repeated cycles
Textbook interpretation	Replenishment can create a relatively steady composition	Hydroperoxides can accumulate during cooling and decompose upon reheating
Practical caution	Fresh-oil addition does not automatically establish safety	Repeated heating is an important deterioration risk

[HPSC EXAM TRAP] The textbook's steady-state explanation is not permission to continue using oil indefinitely. Turnover, deterioration products and applicable standards remain relevant.

9. Factors controlling degradation

- Frying temperature.
- Duration of heating.
- Oxygen exposure and exposed surface area.
- Heat-transfer method.
- Metals contacting the oil.
- Quantity and nature of the food.
- Frying capacity and throughput.
- Oil turnover and replenishment.
- Food moisture and released solids.
- Starting oil composition and antioxidant protection.

10. Total Polar Compounds: interpretation and statutory boundary

What the measurement means

Chemical clarification: Total Polar Compounds represent a broad method-defined group of polar constituents in oil. These can include oxidised lipids, partial glycerides, free fatty acids and some degradation-related products. It is not another name for peroxide value and does not represent one single toxic chemical.

A compositional expression is:

$$Total\ polar\ compounds\ (percent) = \frac{Mass\ of\ polar\ fraction}{Mass\ of\ oil\ sample}\times100$$

Reference fractionation and rapid instrumental estimates are not automatically interchangeable without calibration and validation.

Requested regulatory figures

Requested figure	Evidence status in the supplied corpus
Fresh or unused frying oil: maximum 15%	Not established by either upload
Used frying oil: maximum 25%	Not established by either upload

These figures are therefore **not cited here as verified provisions of the uploaded fortification compendium**. Their exact governing regulation, wording, applicability and boundary rule require the relevant food-products or frying-oil notification.

A maximum stated as “not more than 25%” and a prohibition stated as “25% or more” differ at the exact boundary. Do not decide that point without the actual legal text.

11. Defensible quality-control approach

1. Control time and temperature rather than relying only on appearance.
2. Remove accumulated food debris using appropriate operating procedures.
3. Minimise unnecessary idle heating and repeated thermal cycling.
4. Assess the oil with measurements appropriate to its history.
5. Interpret peroxide value cautiously in heated oil.
6. Apply the verified statutory discard criterion from the governing instrument.
7. Record oil use, replacement and disposal as required by the applicable framework.

Answer-writing conclusion: Deep frying couples heat and mass transfer with hydrolysis, oxidation and polymerisation. Sensory defects are useful warnings, but analytical assessment and verified legal criteria determine a defensible safety decision.

4. Module 6 — Enzymes & Kinetics

MODULE 6: FOOD ENZYMES AND INDUSTRIAL KINETICS

Core source: *Food Chemistry*, Lessons 6, 15 and 16.

6.1. First principle: an enzyme changes reaction speed, not the equilibrium destination

An enzyme provides a lower-energy route to a chemical transformation. It does not supply unlimited energy, make an unfavourable process favourable by itself, or alter the equilibrium constant of the uncoupled reaction.

The textbook treats enzymes as proteinaceous biological catalysts with an active centre. The active site includes regions important for substrate binding and catalysis.

Chemical clarification: Most familiar food-processing enzymes are proteins, but the statement “all biological catalysts are proteins” is not universally true because catalytic ribonucleic acids exist.

6.2. Enzyme architecture

Intuitive opening

Some enzymes are complete as proteins alone. Others need an additional chemical component. Removing that necessary component leaves an incomplete catalytic system.

```
Apoenzyme + required cofactor --> holoenzyme
Protein part + nonprotein component --> complete functional enzyme system
```

Term	Meaning	Important distinction
Apoenzyme	Protein portion of a cofactor-dependent enzyme	Not the same as a zymogen
Cofactor	Required nonprotein component	Can be a metal ion or organic molecule
Prosthetic group	Tightly associated component	Defined primarily by its association with the protein
Coenzyme	Organic catalytic partner	Often loosely associated in the textbook’s terminology
Holoenzyme	Complete enzyme with its required component	Catalytically competent under suitable conditions
Zymogen or proenzyme	Inactive precursor requiring activation	Often activated by limited proteolysis

Examples of activation

```
Pepsinogen --> pepsin
Trypsinogen --> trypsin
```

A zymogen is not necessarily inactive because it lacks a cofactor. Its peptide structure may prevent activity until an activation event occurs.

Chemical clarification: Organic coenzymes can also be tightly bound and function as prosthetic groups. The textbook's loose-versus-tight distinction is useful introductory terminology, not an absolute division of all organic cofactors.

6.3. Six textbook enzyme classes

Intuitive opening

Classify an enzyme by the reaction it catalyses, not by the food in which it is found.

Class	Reaction principle	Textbook example or food-relevant example
Oxidoreductases	Transfer of electrons or hydrogen; oxidation–reduction	Catalase, peroxidase and glucose oxidase
Transferases	Transfer of a functional group between molecules	Glucokinase
Hydrolases	Bond cleavage involving water	Lipase, invertase, proteases and alkaline phosphatase
Lyases	Nonhydrolytic elimination or addition across double bonds	Fumarate hydratase; pectate lyase
Isomerases	Rearrangement within a molecule	Mutases
Ligases	Bond formation coupled to energy-yielding nucleotide cleavage	Enzymes forming new covalent bonds using adenosine triphosphate

The first digit of an Enzyme Commission number identifies the class.

Enzyme	Textbook number	Interpretation
Alkaline phosphatase	3.1.3.1	Hydrolase acting on phosphoric monoesters
Lipase	3.1.1.3	Hydrolase acting on carboxylic ester bonds
Catalase	1.11.1.6	Oxidoreductase

Historical scope: The upload teaches the traditional six-class system. Modern Enzyme Commission classification also includes translocases as a seventh class. For a source-specific question asking the textbook's six classes, reproduce the six listed above; do not assert that the present classification contains only six.

[HPSC EXAM TRAP] "All pectin-degrading enzymes are hydrolases" is false. Pectate lyase belongs to the lyases.

6.4. Michaelis–Menten kinetics from first principles

Intuitive opening

Imagine a fixed number of catalytic workstations. When substrate is scarce, supplying more substrate increases the number of occupied workstations. When nearly every workstation is occupied, additional substrate cannot appreciably increase the rate.

Elementary scheme

```
E + S <--> ES --> E + P
```

Where:

- **E** means free enzyme.
- **S** means substrate.
- **ES** means enzyme–substrate complex.
- **P** means product.
- **k1** is the association rate constant.
- **k−1** is the dissociation rate constant.
- **kcat** is the catalytic conversion rate constant in this simple model.

Conditions for the elementary equation

The derivation assumes initial-rate measurement, negligible reverse product reaction, substrate sufficiently in excess of enzyme, and an approximately steady concentration of the enzyme–substrate complex after the brief initial transient. Not every allosteric or multisubstrate enzyme follows this simple form.

Derivation

Formation of the enzyme–substrate complex approximately equals its disappearance:

$$k_1[E][S] = (k*{-1}*+*k*{cat})[ES]$$

Define:

$$K_m = \frac{k*{-1}*+*k*{cat}}{k_1}$$

The total enzyme concentration is:

$$[E]_T = [E]+[ES]$$

Solving gives:

$$[ES] = \frac{[E]*T[S]}{K_m+[S]}$$*

Because the initial rate is catalytic conversion of the complex:

$$v_0 = k_{cat}[ES]$$

Therefore:

$$v_0 = \frac{V*{max}*[S]}{K_m+[S]}$$*

with:

$$V*{max}* = *k*{cat}[E]_T$$

Evidence note: The upload describes saturation and gives the half-maximum relationship, but does not supply this full steady-state derivation. The derivation corrects its oversimplified description of the Michaelis constant as an equilibrium constant.

The three substrate regimes

Substrate concentration	Approximate expression	Order with respect to substrate
Much less than the Michaelis constant	Initial rate approximately equals maximum rate multiplied by substrate concentration divided by the Michaelis constant	First order
Comparable to the Michaelis constant	Full Michaelis–Menten equation is needed	Mixed or fractional order
Much greater than the Michaelis constant	Initial rate approaches maximum rate	Approximately zero order

Low substrate	Intermediate substrate	High substrate
Many sites empty	Occupancy increases	Sites nearly saturated
Rate tracks supply	Curved response	Rate approaches plateau

Michaelis constant: meaning and units

At substrate concentration equal to the Michaelis constant:

$$v_0 = \frac{V{max}K_m}{K_m+K_m} = \frac{V{max}}{2}$$

Thus the Michaelis constant is the substrate concentration producing half the maximum rate **for the applicable model**. Its units are concentration units, not rate units.

Affinity: the qualified interpretation

The binding dissociation constant is:

$$K_d = \frac{k_{-1}}{k_1}$$

If catalytic conversion is much slower than complex dissociation, the Michaelis constant approaches the dissociation constant. Otherwise, it incorporates catalysis as well as binding.

> **[HPSC EXAM TRAP]** A lower Michaelis constant often suggests greater apparent substrate affinity, but “the Michaelis constant always equals the enzyme–substrate binding constant” is incorrect.

Hypothetical worked rate calculation

Maximum rate: 100 micromoles per minute. Michaelis constant: 2 millimoles per litre.

Substrate concentration, millimoles per litre	Initial rate, micromoles per minute
0.2	9.09
2	50.00
20	90.91

At a substrate concentration ten times the Michaelis constant, the rate is approximately 90.9% of maximum, not exactly 100%.

6.5. Lineweaver–Burk double-reciprocal plot

Intuitive opening

The original saturation curve is curved. Taking reciprocals transforms the elementary equation into a straight-line form.

$$\frac{1}{v_0} = \frac{K_m}{V{max}}\frac{1}{[S]} + \frac{1}{V{max}}$$

Plot feature	Value
Vertical axis	Reciprocal of initial rate
Horizontal axis	Reciprocal of substrate concentration
Slope	Michaelis constant divided by maximum rate
Vertical intercept	Reciprocal of maximum rate
Horizontal intercept	Negative reciprocal of Michaelis constant

Hypothetical example: With maximum rate 100 and Michaelis constant 2 in the preceding units:

- Slope is 0.02 in the corresponding reciprocal-plot units.
- Vertical intercept is 0.01.
- Horizontal intercept is −0.5.

Chemical clarification: The plot magnifies errors at low substrate concentration. It is useful for examination interpretation, but fitting the untransformed rate equation is generally preferable for accurate parameter estimation.

6.6. Reversible inhibition

Intuitive opening

An inhibitor may prevent substrate binding, disable catalytic turnover after binding, or bind preferentially to the enzyme–substrate complex. The kinetic effect depends on which enzyme forms it binds and how strongly.

Type	Binding pattern	Apparent Michaelis constant	Apparent maximum rate	Effect of high substrate
Competitive	Binds free enzyme in a manner mutually exclusive with substrate binding	Increases	Unchanged	Inhibition can be overcome in the elementary model
Pure noncompetitive	Binds free enzyme and enzyme–substrate complex with equal affinity	Unchanged	Decreases	Cannot restore the original maximum rate
Uncompetitive	Binds only enzyme–substrate complex	Decreases	Decreases	Does not overcome inhibition
Mixed	Binds free enzyme and enzyme–substrate complex with unequal affinities	May increase or decrease	Decreases	Does not restore the original maximum rate

General mathematical framework

Let:

$$\alpha = 1+\frac{[I]}{K_i}$$

$$\alpha' = 1+\frac{[I]}{K_i'}$$

Here **I** is inhibitor concentration, **Ki** describes binding to free enzyme, and **Ki′** describes binding to the enzyme–substrate complex.

Then:

$$v_0 = \frac{V*{max}[S]}{\alpha K_m+\alpha'[S]}$$

Consequently:

$$V*{max,app} = \frac{V{max}}{\alpha'}$$

$$K*{m,app} = \frac{\alpha K_m}{\alpha'}$$

Competitive inhibition

For competitive inhibition, alpha-prime equals one.

- Apparent Michaelis constant increases by the factor alpha.
- Maximum rate stays unchanged.
- Reciprocal-plot lines share the vertical intercept.
- Slope increases.
- Horizontal intercept moves toward zero.

Textbook example: Malonate competitively inhibits succinate dehydrogenase in the succinate comparison.

Pure noncompetitive inhibition

For pure noncompetitive inhibition, alpha equals alpha-prime.

- Maximum rate decreases.
- Michaelis constant remains unchanged.
- Reciprocal-plot lines share the horizontal intercept.
- Vertical intercept and slope increase.

Binding away from the substrate-binding site is not enough, by itself, to prove these exact kinetic changes.

Uncompetitive inhibition

For uncompetitive inhibition, alpha equals one and alpha-prime exceeds one.

- Maximum rate decreases.
- Michaelis constant decreases by the same factor.
- Their ratio, and therefore reciprocal-plot slope, remains unchanged.
- Reciprocal-plot lines are parallel.

A lower apparent Michaelis constant does not mean the inhibited enzyme processes substrate more successfully; the maximum rate is also lower.

Hypothetical numerical comparison

Starting values: Michaelis constant 2 millimoles per litre; maximum rate 100 micromoles per minute. Use an inhibition factor of two.

Condition	Apparent Michaelis constant	Apparent maximum rate
No inhibitor	2	100
Competitive	4	100
Pure noncompetitive	2	50
Uncompetitive	1	50

[HPSC EXAM TRAP] Competitive inhibition changes the **apparent** Michaelis constant. Do not claim that the inhibitor necessarily changes the intrinsic binding chemistry of every remaining unbound enzyme molecule.

6.7. Irreversible inhibition

The textbook emphasises covalent attachment to an enzyme group and failure to restore activity through dilution or dialysis.

Chemical clarification: Some effectively irreversible inhibitors act through exceptionally tight binding rather than a covalent bond. The operational feature is persistent loss of activity on the relevant timescale.

```
Active enzyme + irreversible inhibitor --> persistently inactive enzyme
```

Loss of active enzyme commonly lowers the observed maximum rate. The full kinetic pattern depends on mechanism and exposure history; irreversible inhibition should not automatically be equated with pure noncompetitive inhibition.

Feature	Reversible inhibition	Irreversible inhibition
Binding or reaction	Dynamic binding equilibrium	Persistent chemical modification or effectively irreversible association
Dilution	May reduce inhibition	Often does not restore activity
Time dependence	Depends on model and equilibration	Often increases with exposure time

Feature	Reversible inhibition	Irreversible inhibition
Enzyme recovery	Possible after removing inhibitor	Usually requires new enzyme or a specific reactivation mechanism

6.8. Other factors affecting enzyme activity

Factor	First-principles explanation	Exam-safe qualification
Enzyme concentration	More active sites raise capacity when substrate is available	Initial rate is commonly proportional to active enzyme concentration under controlled conditions
Temperature	Reaction acceleration competes with denaturation	Optimum temperature depends on exposure time, substrate and enzyme source
Hydrogen ion concentration	Alters ionisation of catalytic groups and substrate, and can alter structure	Optimum hydrogen ion concentration need not equal the isoelectric point
Activators	Required ions or cofactors support catalysis	Examples must be enzyme-specific
Product accumulation	Can reduce net forward rate or inhibit enzyme	Initial-rate measurements minimise this complication

Correction of textbook absolutes: Not all enzymes stop at zero degrees Celsius, not all denature near fifty degrees Celsius, and not all function best near neutrality. Thermostable enzymes and enzymes adapted to acidic or alkaline conditions exist.

6.9. Industrial and endogenous food enzymes

A. Invertase

Name: Beta-D-fructofuranosidase.

Sucrose + water --> glucose + fructose

The equimolar mixture is invert sugar. The name inversion refers to the change in optical rotation, not merely to increased sweetness.

Application: Confectionery, including softening or liquefaction of sucrose-rich centres and reduction of sucrose crystallisation tendency.

B. Pectinolytic enzymes

Enzyme	Main action	Product or processing implication
Pectin methylesterase	Hydrolyses methyl ester groups	Produces methanol and demethylated carboxyl groups; calcium-mediated flocculation may follow
Polygalacturonase	Hydrolyses alpha-1,4 linkages in the galacturonic-acid chain	Depolymerises suitable pectic substrates and reduces viscosity
Pectate lyase	Cleaves by beta-elimination rather than hydrolysis	Forms unsaturated products from suitable pectate substrates

Application: Improved extraction, viscosity reduction and clarification of fruit and vegetable juices.

Qualification: Controlled pectin modification can also promote firmness under some conditions. Every pectinolytic action does not inevitably produce the same texture outcome.

C. Protease architecture and applications

Group	Defining feature in the textbook	Examples
Acid proteases	Active under acidic conditions	Pepsin and chymosin
Serine proteases	Catalytic serine-containing active site	Trypsin, chymotrypsin and plasmin
Sulphydryl or cysteine proteases	Essential cysteine-associated catalytic chemistry	Papain, ficin and bromelain
Metal-dependent proteases	Require a catalytic metal	Carboxypeptidases and aminopeptidases

Chemical clarification: Metal-dependent proteases are not all exopeptidases; the textbook's generalisation is too narrow.

Chymosin in cheese manufacture

```
Kappa-casein
  |
  | Chymosin cleaves the phenylalanine-105 / methionine-106 bond
  v
Para-kappa-casein + soluble glycomacropeptide
  |
  | Destabilised micelles aggregate in suitable calcium conditions
  v
Cheese curd
```

The first stage is enzymatic cleavage. The second stage is aggregation of modified casein micelles.

Terminology trap: Rennin is the older name for chymosin. It is not **renin**, the enzyme of the blood-pressure regulatory system.

Plant proteases

Enzyme	Common plant association	Applications discussed in the textbook
Papain	Papaya	Meat tenderisation and beer-haze reduction
Ficin	Fig	Meat tenderisation and beer-haze reduction
Bromelain	Pineapple	Meat tenderisation and beer-haze reduction

Beer haze can involve interactions between polypeptides and tannin-related compounds. Reducing peptide size can decrease the ability to form visible haze particles.

D. Other useful enzyme links

- **Lactase:** Converts lactose into glucose and galactose; useful in lactose-reduced dairy products.
- **Lipase:** Produces controlled cheese flavour but can cause hydrolytic rancidity.
- **Alpha-amylase:** Random cleavage of alpha-1,4 starch linkages; useful in liquefaction and baking.
- **Beta-amylase:** Releases maltose units from nonreducing ends.
- **Pullulanase:** Cleaves alpha-1,6 branch linkages in suitable substrates.

- **Glucose oxidase:** Removes glucose or oxygen through oxidation; catalase may be used to remove the hydrogen peroxide generated.
- **Polyphenol oxidase:** Causes undesirable browning in damaged produce but contributes to desirable tea oxidation.

6.10. Blanching and pasteurisation indicators

Intuitive opening

An indicator enzyme is selected because its survival or inactivation provides information about the adequacy of a process. The test does not directly count every pathogen or prove sterility.

Process	Indicator	Expected satisfactory result	Interpretation
Vegetable blanching	Peroxidase	No detectable residual activity under the prescribed test	Evidence of sufficient enzyme-inactivation treatment
Milk pasteurisation	Alkaline phosphatase	Negative prescribed activity test	Evidence supporting adequate pasteurisation and absence of relevant raw-milk contamination

Peroxidase reaction

Hydrogen peroxide + reduced donor --> water + oxidised donor

Peroxidase is relatively heat resistant among important vegetable enzymes and is therefore commonly used for blanching assessment.

Qualification: Do not call it the most heat-stable enzyme in every organism and every food. Indicator selection and residual-activity criteria are process-specific.

Alkaline phosphatase

Phosphate monoester + water --> alcohol + inorganic phosphate

The textbook identifies absence of alkaline phosphatase activity as a pasteurisation indicator.

Chemical clarification: Alkaline phosphatase is not lactoperoxidase. Milk lactoperoxidase is a haem-containing enzyme, despite the upload's erroneous description of it as a flavoprotein.

> **[HPSC EXAM TRAP]** A negative phosphatase test is not a declaration that milk is sterile. Pasteurisation and sterilisation are different processes.

5. Module 7 — Natural Pigments

MODULE 7A: NATURAL FOOD PIGMENTS

Evidence boundary

The uploaded textbook directly treats anthocyanins, carotenoids, annatto and betalains in Lesson 20, and mentions chlorophyll degradation during ripening and tea processing. It does not provide a complete structural account of chlorophyll derivatives, anthoxanthins or meat pigments. Those requested topics are identified below as chemical clarifications rather than attributed to the missing Srilakshmi text.

7.1. First principle: colour comes from molecular electron behaviour

Pigments absorb selected wavelengths of visible light. Conjugated systems, metal coordination and changes in molecular form determine the light absorbed and therefore the colour observed.

A small chemical change can therefore create a large visible change. Replacing a coordinated metal, changing hydrogen ion concentration or oxidising iron may alter colour without removing the entire pigment molecule.

7.2. Overview

Pigment family	Principal structural feature	General solubility tendency	Representative colours
Chlorophylls	Magnesium-containing tetrapyrrole system with a phytol-associated ester	Lipid-associated	Green
Carotenoids	Extended conjugated polyene system	Usually lipid-associated; exceptions and formulated dispersions exist	Yellow, orange and red
Anthocyanins	Glycosylated flavonoid pigments	Water-soluble	Red, purple and blue
Anthoxanthins	Pale flavonoid pigments	Frequently associated with aqueous plant systems	White, cream or pale yellow
Betalains	Nitrogen-containing pigments	Water-soluble	Red-violet or yellow-orange
Myoglobin derivatives	Haem iron within a protein environment	Protein-associated tissue pigment	Purple-red, bright red, brown and cured pink

7.3. Chlorophylls

Intuitive opening

Chlorophyll resembles a coloured molecular ring holding magnesium at its centre, with a long hydrophobic tail anchoring it in the plant membrane environment.

Chemical clarification: The common introductory description is a porphyrin-like tetrapyrrole ring. More precisely, major chlorophylls have a **chlorin** macrocycle, not an unchanged ordinary porphyrin ring.

Chlorophyll a and chlorophyll b

They share the magnesium-containing framework but differ in a substituent: chlorophyll a has a methyl group where chlorophyll b has a formyl group. This affects spectral behaviour and green shade.

Acid conversion to pheophytin

Acid replaces central magnesium with hydrogen ions.

```
Chlorophyll + acid
      |
      | Magnesium displaced
      v
Pheophytin + released magnesium
Green --> olive-brown
```

Heat and acid exposure accelerate the visible loss of fresh green colour.

Dephytylation and demetallation are separate events

Pigment	Central magnesium	Phytol tail	Teaching significance
Chlorophyll	Present	Present	Native green pigment
Chlorophyllide	Present	Removed	Tail cleavage without magnesium loss
Pheophytin	Removed	Present	Acid-associated olive-brown pigment
Pheophorbide	Removed	Removed	Both transformations have occurred

Alkali and chlorophyllin

Alkaline treatment can hydrolyse ester groups and produce more water-compatible chlorophyllin derivatives or salts, often associated with a brighter green appearance.

Qualification: “Alkali always converts all chlorophyll into one single stable bright-green molecule” is an oversimplification. The products depend on conditions.

Adding alkali during cooking may adversely affect texture and nutrient retention. A visually green product is not automatically nutritionally superior.

Copper complexes

Copper-containing chlorophyll derivatives can be more colour-stable than the corresponding magnesium-containing system under some processing conditions.

Legal boundary: A chemical ability to stabilise colour does not authorise adding an unapproved copper salt to food. The permitted additive form and food category must be checked separately.

[HPSC EXAM TRAP] Pheophytin formation means magnesium removal. It does not necessarily mean phytol removal.

7.4. Carotenoids

Intuitive opening

A long chain of alternating single and double bonds allows electrons to respond to visible light. This extended conjugation explains carotenoid colour and also creates sites vulnerable to oxidation.

Uploaded-source finding: Parent carotenoids commonly contain forty carbon atoms assembled from eight isoprene units.

Carotenes versus xanthophylls

Group	Chemical distinction	Examples
Carotenes	Hydrocarbon carotenoids	Beta-carotene and lycopene
Xanthophylls	Oxygen-containing carotenoids	Lutein, zeaxanthin and canthaxanthin

Canthaxanthin is a diketo carotenoid. Oxygen may occur in hydroxyl, epoxy or carbonyl groups in different xanthophylls.

Vitamin A activity

Beta-carotene is a provitamin A carotenoid. Lycopene, lutein and zeaxanthin should not be assigned vitamin A activity merely because they are carotenoids.

Processing responses

- Oxidation can bleach colour by disrupting conjugation.
- Heat and light can promote isomerisation and degradation.
- Food structure and formulation affect apparent colour and pigment availability.
- Carotenoids are often less directly responsive to ordinary food hydrogen ion concentration changes than anthocyanins, but they are not chemically indestructible.

Annatto: bixin and norbixin

Uploaded-source finding: Annatto originates from the orange-red coating of *Bixa orellana* seeds. The textbook distinguishes oil-soluble and water-soluble preparations and notes their uses in butter, margarine, cheese and other products.

Chemical clarification: Bixin and norbixin are oxygen-containing **apocarotenoids**, formed from a shortened carotenoid framework. Bixin contains twenty-five carbon atoms; norbixin contains twenty-four. They should not be forced into an absolute statement that every carotenoid-related pigment is a forty-carbon xanthophyll.

Pigment	Practical distinction
Bixin	Relatively lipid-compatible; methyl ester-related annatto pigment
Norbixin	Dicarboxylic acid-related pigment; its salts support aqueous preparations

```
Bixin ester group + hydrolytic treatment --> norbixin-related form
                                  |
                                  +--> Greater aqueous compatibility as salts
```

Numerical textbook examples: not legal permissions

The textbook gives product-use illustrations for beta-carotene, canthaxanthin and annatto. These descriptive figures must not be substituted for current permitted category limits.

[HPSC EXAM TRAP] “All carotenoids are provitamin A compounds” and “all carotenoid derivatives contain forty carbon atoms” are both incorrect.

7.5. Anthocyanins

Intuitive opening

Anthocyanins can exist in several interconverting molecular forms. Hydrogen ion concentration changes the balance among these forms, producing different colours.

Uploaded-source finding: They are water-soluble pigments associated with red, violet and blue colours in fruits and vegetables, including grapes, berries, red cabbage and brinjal.

Chemical clarification: An anthocyanin is generally a sugar-containing derivative of an anthocyanidin. The glycoside and aglycone are not interchangeable terms.

Hydrogen ion concentration and molecular form

Environment	General tendency	Necessary qualification
Strongly acidic	Red flavylium-cation form is favoured	Exact shade depends on pigment and matrix
Less acidic to near-neutral	Purple or blue forms may contribute	Colourless hydrated forms can also be substantial
Alkaline	Blue-green shifts may occur transiently	Instability and irreversible degradation can follow

```
Red flavylium-associated form
        <--> hydrated and other interconverting forms
        <--> quinonoidal forms
                    |
                    +--> Colour depends on conditions and composition
```

The correct model is an equilibrium network, not a universally reversible red-to-blue switch.

Sulphur dioxide bleaching

Sulphur dioxide-derived bisulphite can form colourless addition products with anthocyanin structures, reducing visible colour.

Evidence boundary: This molecular explanation is chemical clarification; the uploaded pigment section does not develop the full bleaching mechanism.

Other determinants

Temperature, oxygen, light, metal ions, copigmentation, acylation and the food matrix affect colour retention.

[HPSC EXAM TRAP] Blue colour at high hydrogen ion concentration is not the expected simple anthocyanin rule. Acidic conditions generally favour the red flavylium form.

7.6. Anthoxanthins

Source gap: The supplied textbook does not provide the requested dedicated anthoxanthin treatment.

Chemical clarification: Anthoxanthin is a traditional food-science term for pale flavonoid pigments, including certain flavones and flavonols. They commonly contribute white, cream and pale-yellow appearances.

- Acidic conditions often favour a pale appearance.
- Alkaline conditions can intensify yellow colour.
- Complex formation with metals can change shade.

These tendencies depend on the individual compound; the term does not identify one universal pigment molecule.

7.7. Betalains

Intuitive opening

Beetroot red is not simply another anthocyanin red. Betalains contain nitrogen and have a different biosynthetic and structural foundation.

Group	Typical colour	Example association
Betacyanins	Red to violet	Betanin in red beetroot
Betaxanthins	Yellow to orange	Yellow betalain pigments

Uploaded-source finding: The textbook identifies betalains as water-soluble, nitrogen-containing pigments and describes their food-colour applications.

Chemical correction: Its description of betalains as an indigoid group is outdated or inaccurate. They are better described as tyrosine-derived pigments based on betalamic acid chemistry. This distinction matters because **indigo carmine** is an indigoid synthetic dye, while beetroot betalains are not the same chemical family.

Processing behaviour

Heat, oxygen and other environmental conditions can degrade betalain colour. Their response to hydrogen ion concentration is not the same anthocyanin flavylium equilibrium.

> **[HPSC EXAM TRAP]** “Betacyanins are blue because their name contains cyan” is incorrect. They are red-violet pigments.

7.8. Meat pigments

Source gap: The uploaded textbook does not provide a dedicated myoglobin-conversion section. The following is the chemical clarification requested for answer-writing completeness.

Intuitive opening

Meat colour depends on the oxidation state of haem iron, the ligand attached to it and the condition of the surrounding protein. Oxygen binding and iron oxidation are separate events.

Pigment form	Iron state	Ligand or condition	Typical appearance
Deoxymyoglobin	Ferrous, iron(II)	No oxygen bound at the relevant site	Purplish red
Oxymyoglobin	Ferrous, iron(II)	Oxygen bound	Bright red
Metmyoglobin	Ferric, iron(III)	Oxidised haem environment	Brown
Nitrosylmyoglobin	Predominantly ferrous haem with nitric oxide	Nitric oxide binding during curing	Red cured precursor
Nitrosohemochrome	Nitric-oxide-associated haem in heat-denatured protein environment	Heating of the cured system	Cured pink

Correct pathway architecture

```
Deoxymyoglobin <--> Oxymyoglobin
   Fe(II)            Fe(II)
     \                /
      \  Oxidation   /
```

```
      v          v
    Metmyoglobin
      Fe(III)
        |
        | Reduction can restore a ferrous form
        v
    Ferrous myoglobin-related forms

Curing branch:
Suitable ferrous myoglobin + nitric oxide --> nitrosylmyoglobin
                          |
                          | Heating and protein denaturation
                          v
                  nitrosohemochrome
                  cooked cured pink
```

Do not memorise the whole system as an obligatory linear sequence from deoxymyoglobin through metmyoglobin directly to nitrosohemochrome.

Descriptive-answer distinctions

- Oxygenation does not itself change iron(II) to iron(III).
- Metmyoglobin formation is oxidation.
- Cured pink requires nitric-oxide-related chemistry and the relevant heat treatment.
- Brown meat colour alone is not a complete microbiological safety diagnosis.

7.9. Comparative processing table

Treatment	Chlorophyll	Carotenoids	Anthocyanins	Betalains	Myoglobin system
Acid	Magnesium loss can produce pheophytin	Often less direct colour-equilibrium effect	Red flavylium form generally favoured	Stability depends on conditions	Protein and haem environment can change
Alkali	Hydrolysis and chlorophyllin-related changes	Not a universal protection method	Colour shift followed by possible degradation	Not governed by the anthocyanin mechanism	Not a general colour-preservation strategy
Oxygen	Can contribute to degradation	Oxidation can bleach conjugated colour	Can contribute to oxidative losses	Can contribute to degradation	Oxygenation and oxidation must be distinguished
Heat	Accelerates pigment transformations	Can cause isomerisation and degradation	Often reduces stability	Often reduces stability	Protein denaturation and cooked-pigment formation

Answer-writing conclusion: Natural pigment behaviour is structure-specific. A scientifically strong answer identifies the chromophore, states the chemical transformation and connects it to the observed colour without confusing colour stability with legal permission or food safety.

6. Flavour, Sweeteners & Colours

MODULE 7B: FLAVOUR ENHANCERS, INTENSE SWEETENERS AND FOOD COLOURS

Core source: *Food Chemistry*, Lessons 18–20, printed pages 99–110.

1. First principle: flavour is a combined sensory experience

Taste is largely associated with compounds interacting with oral receptors. Aroma is associated with volatile compounds reaching the olfactory system. Texture also contributes to the overall experience called flavour.

A flavour enhancer changes the perceived flavour of a food. It need not be chemically or sensorially identical to an aroma compound.

Odour threshold and aroma contribution

The textbook defines an aroma contribution ratio as the concentration of a compound divided by its odour threshold in the food.

$$Aroma\ value = \frac{Concentration\ in\ food}{Odour\ threshold\ in\ that\ food}$$

A value above one supports the possibility of a detectable contribution, but interactions among compounds and the matrix still matter.

2. Monosodium glutamate

Intuitive opening

Glutamate provides a savoury taste rather than a fruity aroma. Its effect is especially useful in suitable soups, meat, vegetable and seafood systems.

Uploaded-source findings:

- Monosodium glutamate is associated with umami taste.
- The L-form is the relevant flavour-active form in the textbook account.
- Pure monosodium glutamate is odourless.
- The textbook describes stronger effects around hydrogen ion concentration conditions corresponding to a pH of six to eight, with reduced effect at lower pH.

Chemical clarification: Glutamate activates umami-related taste receptors. Describing it as having no taste of its own is incompatible with recognising umami as a primary taste.

Synergy with 5-prime ribonucleotides

Compound	Main teaching association
5-prime inosinate	Umami enhancement and synergy with glutamate
5-prime guanylate	Umami enhancement and synergy with glutamate

```
L-glutamate + suitable 5-prime ribonucleotide
                  |
                  v
Greater perceived umami than a simple independent-addition model predicts
```

The position of the phosphate matters. Do not assume that every nucleotide or every positional isomer has the same flavour-enhancing activity.

Safety interpretation

The textbook discusses historically reported symptoms and the term “Chinese restaurant syndrome.” That historical passage should not be converted into a claim that normal use reliably causes a defined disease in every consumer. Safety assessment, individual responses and legal category permissions are separate questions.

[HPSC EXAM TRAP] “Monosodium glutamate is an aroma compound because it enhances flavour” is incorrect. Its central role is a taste effect.

3. Maltol and ethyl maltol

Intuitive opening

A caramel-like aroma can increase the perception of sweetness without acting as a direct gram-for-gram replacement for sugar.

Compound	Structural description	Sensory function
Maltol	3-hydroxy-2-methyl-4-pyrone	Caramel-like odour; enhances perceived sweetness and can mask bitterness
Ethyl maltol	Corresponding ethyl-substituted pyrone	Similar aroma enhancement, stronger in the textbook comparison

Uploaded-source finding: Ethyl maltol is described as approximately four to six times more powerful than maltol.

The textbook gives illustrative maltol use ranges and sweetness-perception effects. These are not a substitute for the relevant food-additive permission and category limit.

Distinction: A flavour modifier that enhances sweetness perception is not automatically an intense sweetener with the same receptor action or technological role as sucralose.

4. Bulk versus intense sweeteners

Intuitive opening

Sugar supplies sweetness and physical material. A tiny amount of an intense sweetener can replace much of the sweetness but not necessarily the bulk, viscosity, browning behaviour or preservation effect.

Feature	Bulk sweetener	Intense sweetener
Typical amount needed	Relatively large	Relatively small
Physical body	Commonly substantial	Usually limited at normal use levels
Energy	Often appreciable, depending on substance	Usually small per equivalent sweetness
Technological equivalence to sucrose	Partial or substantial depending on compound	Usually incomplete
Metabolism	Substance-specific	Also substance-specific

Polyols

The textbook includes sorbitol, mannitol and other sugar alcohols. Sorbitol can be produced by hydrogenating glucose and is hygroscopic.

Qualification: The broad textbook claim that all bulk sweeteners are completely metabolised and all intense sweeteners are excreted unchanged is not chemically reliable. Classification by sweetness intensity and classification by metabolism are different.

5. Requested intense sweeteners

Sweetener	Approximate sweetness compared with sucrose	Chemical or processing distinction	Evidence status
Saccharin	About 300 times	Orthobenzoic sulphimide-related sweetener; sodium salt is highly soluble; bitter aftertaste can occur at higher concentration	Directly described in upload
Aspartame	Commonly about 200 times	Methyl ester of an aspartic-acid–phenylalanine dipeptide; metabolised; unsuitable for people with phenylketonuria	Named in upload; detailed chemistry is clarification
Acesulfame potassium	About 200 times	High-intensity sweetener with useful heat stability and blending synergy	Directly described in upload
Sucralose	About 600 times	Chlorinated sucrose derivative	Directly described in upload

Saccharin

The sodium and calcium salts are discussed in the textbook. Sodium saccharin is widely useful because of solubility and stability.

Aspartame

Chemical clarification: Aspartame provides energy per gram when metabolised, but its high sweetness means a much smaller quantity is used than sucrose. It is therefore incorrect to infer zero metabolism from the label intense sweetener.

Its phenylalanine component creates a specific concern for people with phenylketonuria. Heat and prolonged storage in some aqueous conditions can reduce its useful sweetness.

Acesulfame potassium

The textbook describes stability under normal pasteurisation conditions and synergy with other sweeteners. This does not mean unlimited stability under every temperature, exposure duration and food condition.

Sucralose

Chemical correction: Three hydroxyl groups in the sucrose-related structure are replaced by **chlorine atoms**, not by chlorine molecules.

Sucralose is not sucrose simply mixed with chloride salt. Its chemical structure is changed.

Do not reproduce unreliable safety figures

The upload contains problematic acceptable-daily-intake figures, including an apparent gram-versus-milligram error for sucralose. This handout does not reproduce them as authoritative safety limits.

Source gap: Current acceptable daily intakes, category permissions and mandatory sweetener declarations require the relevant authoritative safety and regulatory documents.

[HPSC EXAM TRAP] Relative sweetness, acceptable daily intake and maximum permitted concentration are three different quantities with different purposes and units.

6. Natural, nature-identical and artificial colours

Category	Meaning	Example
Natural colour	Obtained from a natural source	Annatto or a suitable plant pigment extract
Nature-identical colour	Synthesised compound chemically matching a naturally occurring one	Certain carotenoid preparations
Artificial colour	Synthetic colour not described simply as a naturally extracted pigment	Listed synthetic food dyes
Dye	Produces colour in solution	Water-soluble synthetic dye
Lake	Insoluble colour preparation formed from a dye on a substrate	Aluminium lake

Natural origin does not automatically establish legal permission, unlimited use or absence of adverse effects.

7. Eight synthetic colours listed in the textbook

Intuitive opening

A permitted-colour list is an identity filter. It does not, by itself, say that every listed colour can be added to every food at any concentration.

Colour group	Name	Colour Index number in the upload	Chemical class in the upload
Red	Ponceau 4R	16255	Azo
Red	Carmoisine	14720	Azo
Red	Erythrosine	45430	Xanthene
Yellow	Tartrazine	19140	Pyrazolone
Yellow	Sunset Yellow FCF	15985	Azo
Blue	Indigo Carmine	73015	Indigoid
Blue	Brilliant Blue FCF	42090	Triarylmethane
Green	Fast Green FCF	42053	Triarylmethane

Counting aid: Three reds + two yellows + two blues + one green = **eight**.

The name suffix **FCF** is retained because it is part of the standard colour name.

Chemical clarification: Tartrazine is also structurally an azo dye with a pyrazolone-related framework. If asked to reproduce the uploaded table's class, write pyrazolone; do not conclude that the table's categories are a complete modern structural taxonomy.

Authority boundary

The textbook reproduces an FSSA-labelled list and also contains references to the older Prevention of Food Adulteration framework. It is not a substitute for checking the applicable dated food-additive regulation.

8. Requested general concentration limits

Requested proposition	Source-audited treatment
Maximum 100 milligrams per kilogram for permitted synthetic colours	Not established by the two uploaded documents
Maximum 200 milligrams per kilogram in specified canned products	Exact products and conditions are not established by the uploads

Do not write “all canned products may contain 200 milligrams per kilogram.” An exception must be tied to the precise listed product and provision. Whether a limit applies to a single colour or the total mixture also requires the governing wording.

9. Specific lake-colour example actually present in the upload

The textbook gives the following provision for an aluminium lake of Sunset Yellow FCF in powdered dry beverage mix:

Parameter	Textbook figure
Maximum lake preparation in powdered mix	0.04% by mass
Maximum colour in final reconstituted beverage	8.3 parts per million
Maximum aluminium in final reconstituted beverage	4.4 parts per million

It also requires clear reconstitution instructions in the reproduced provision.

Scope: These are textbook-reproduced figures, not independently verified current law. The concentration in the dry concentrate and the concentration in the beverage as consumed are different bases.

10. Inorganic colours and the titanium dioxide distinction

Uploaded textbook statement: Inorganic colouring matters and pigments are prohibited in its reproduced general provision, with food-grade titanium dioxide allowed in chewing gum up to one percent.

Uploaded fortification finding: Schedule I separately permits titanium dioxide as an ingredient in double fortified salt, together with other listed food-grade substances.

Correct legal reasoning

1. Identify whether the substance is being discussed as a general colour additive or as a component permitted under a specific food standard.
2. Apply the relevant product provision.
3. Do not turn a prohibition on inorganic **colouring matters** into a prohibition on every inorganic nutrient, additive or food constituent.
4. Do not transfer the chewing-gum one-percent figure to double fortified salt without an express provision.

[HPSC EXAM TRAP] “Titanium dioxide is permitted only in chewing gum under all uploaded regulations” is contradicted by the supplied double-fortified-salt schedule.

11. Short answer template: colours and enforcement

Opening: Food colours improve appearance, standardisation and product identity, but their use is controlled by identity, food category, concentration and labelling requirements.

Body:

1. Classify natural and synthetic colours.
2. List the eight textbook synthetic colours by shade.
3. Distinguish dyes from lakes.
4. State the precise category limit only from an available governing provision.
5. Explain why product-specific exceptions cannot be generalised.

Conclusion: Legal permission is a combination of the correct substance, permitted food, permitted use level and applicable declaration—not merely inclusion on a colour list.

7. Module 8 — Fortification Law

MODULE 8: FORTIFICATION REGULATIONS — SOURCE-VERIFIED STATUTORY TABLES

Authority used: Uploaded *Food Safety and Standards (Fortification of Foods) Regulations, 2018*, Version III, dated 30 September 2021.

Temporal boundary: This chapter explains that compendium and its included amendments. It does not certify the absence of subsequent amendments.

8.1. First principle: fortification is a controlled public-health intervention

Adding a nutrient is not enough. A defensible fortification system requires an appropriate food vehicle, suitable nutrient source, controlled concentration, uniform distribution, retention during processing and storage, and compliance with labelling and quality-assurance obligations.

```
Demonstrated nutritional need
        |
        v
Appropriate staple or eligible processed-food vehicle
        |
        v
Permitted nutrient source + specified concentration
        |
        v
Manufacturing control + testing + suitable packaging
        |
        v
Compliant fortified food with required declaration
```

8.2. Legal timeline in the uploaded compendium

Event	Date or requirement
Principal regulations	2018
Original compliance deadline	1 January 2019
Processed-food amendment referenced	18 December 2020
August amendment in force	27 August 2021
Compliance date for the stated August amendments	1 March 2022
September amendment in force	22 September 2021
Compliance date for the stated September amendment	1 April 2022

The original commencement and the transitional compliance dates of later amendments are not the same legal event.

8.3. Definitions under Regulation 2

Fortification

Deliberately increasing the content of essential micronutrients in food to improve its nutritional quality and provide a public-health benefit with minimal risk to health.

Fortificant

A substance added to food to provide micronutrients; the definition excludes nutraceuticals and foods for Special Dietary Uses.

Fortified food

A food specified under the Food Safety and Standards (Food Products Standards and Food Additives) Regulations, 2011, that has undergone fortification according to these regulations.

Fortified Processed Foods

Processed foods altered from their natural state by industrial processing methods and fortified according to these regulations. They may use fortified staples as raw materials and/or be fortified with permitted micronutrients and additives under the referenced food-products and additives framework.

Micronutrients

Essential dietary nutrients, including vitamins, minerals and trace elements, required in very small quantities and important for development, disease prevention and well-being.

Staple foods

Foods intended for mass consumption on a daily basis, including rice, wheat, wheat flour, atta, maida, oil, salt, milk and other foods designated under the regulations.

Government-funded programme

A programme, policy, scheme or other provision through which Central or State Governments sell, distribute or otherwise make food available to the public.

Quality assurance

Systematic measures taken by manufacturers and packers throughout manufacturing or packing to ensure that the finished food complies with the Act and applicable standards and regulations.

Terminology correction: The upload defines staple foods and fortified food separately. Do not invent a distinct quoted statutory definition of “fortified staple foods” where the supplied definition list does not provide one.

8.4. General principles under Regulation 3

Micronutrients may be appropriately added for:

1. Preventing, reducing the risk of, or correcting a demonstrated deficiency.
2. Reducing the risk of, or correcting, inadequate nutritional status.
3. Meeting requirements or recommended intakes.
4. Maintaining or improving health.
5. Maintaining or improving nutritional quality.

Mandatory fortification

Where fortification is made mandatory, it must be based on the severity and extent of public-health need demonstrated through generally accepted scientific evidence.

The Food Authority may specify mandatory fortification of a staple food on the direction of the Government of India.

Important: "When fortified, shall contain" does not mean that every unit of that food is universally required to be fortified by this wording alone.

Heme iron prohibition

Where iron as an iron nutrient source is used, **heme iron shall not be used in any form in any article of food** under the stated provision.

Do not confuse a substance's biological presence in ordinary meat with permission to use it as a fortificant.

Fortified Processed Foods

- May be prepared from fortified cereals and/or milk.
- Shall provide 15–30% of the Indian adult recommended dietary allowance of micronutrients on the stated basis of approximately 600 kilocalories from processed foods, representing about one-third of adult daily energy requirements.
- High Fat Sugar Salt foods are excluded from the **Fortified Processed Foods category**.
- Their applicable micronutrient levels are in Schedule III.

[HPSC EXAM TRAP] The exclusion is category-specific. Do not rewrite it as "no food containing any fat, sugar or salt can ever be fortified."

8.5. Compliance and tolerance under Regulation 4

The September 2021 amendment substitutes the earlier minimum-only formulation with requirements that:

- Added micronutrients declared on the label fall within the Schedule I range.
- A tolerance of minus ten percent of the declared total value may be allowed for compliance and analysis at any point in time.
- The maximum added-micronutrient limit in Schedule I must still be complied with.

What the tolerance does not mean

- It is not an automatic plus-or-minus ten-percent allowance.
- It does not authorise declaring an out-of-range added nutrient value.
- It does not authorise exceeding the statutory maximum through an overage.
- The compendium's wording distinguishes **added micronutrients** from the **declared total value**; preserve that distinction.

Hypothetical tolerance arithmetic

If a relevant declared total nutrient value is 10 units:

$$10\times(1-0.10)=9\ units$$

Nine units represents the numerical minus-ten-percent point. Whether the sample complies still depends on the applicable range, nutrient basis and the complete provision. Arithmetic alone cannot override the maximum limit.

8.6. Schedule I: salt

Intuitive opening

Salt is consumed widely in small daily amounts. Iodine fortification uses this distribution pattern, while double fortified salt adds iron as a second nutrient.

Food vehicle	Nutrient	Level	Permitted source in the upload
Iodized salt	Iodine	15–30 parts per million, dry-weight basis, under the August 2021 substitution	Potassium iodate

Food vehicle	Nutrient	Level	Permitted source in the upload
Double fortified salt	Iodine	15–30 parts per million, dry-weight basis	Potassium iodate
Double fortified salt	Iron, expressed as iron	850–1100 parts per million	Ferrous sulphate or ferrous fumarate

Manufacture versus retail trap

The original table distinguished 20–30 parts per million at manufacture from 15–30 through distribution and retail. The included August amendment substitutes **15–30 parts per million** on a dry-weight basis.

Units

On a mass basis:

$$1\ part\ per\ million = 1\ milligram\ per\ kilogram$$

Thus double fortified salt's iron range corresponds numerically to 850–1100 milligrams of elemental iron per kilogram. It does not mean the same mass of ferrous sulphate or ferrous fumarate preparation.

Additional salt requirements in the schedule notes

- Where an anticaking agent has been added, total water-insoluble matter must not exceed 2.2%.
- Sodium chloride on a dry basis must be at least 97.0% by mass in the stated note.
- Other referenced salt parameters also apply.
- Double fortified salt may contain food-grade hydroxypropyl methylcellulose, titanium dioxide, fully hydrogenated soybean oil and sodium hexametaphosphate.
- Its anticaking agent must not exceed 2.0% on a dry-weight basis.
- Where anticaking agent is used, water-insoluble matter must not exceed 2.2%.

8.7. Schedule I: vegetable oil

Nutrient	Required range when fortified	Permitted source
Vitamin A	6–9.9 micrograms retinol equivalent per gram of oil	Retinyl acetate or retinyl palmitate
Vitamin D	0.11–0.16 microgram per gram of oil	Cholecalciferol or ergocalciferol, only from plant source

Unit conversions for revision

Nutrient	Equivalent per kilogram	International-unit equivalent per gram
Vitamin A	6–9.9 milligrams retinol equivalent	20–33 international units
Vitamin D	110–160 micrograms	4.4–6.4 international units

The schedule specifies:

- One international unit of retinol equals 0.3 microgram retinol equivalent.
- One international unit of vitamin D equals 0.025 microgram.

The legal table is written **per gram of oil**, not per litre and not per kilogram.

8.8. Schedule I: milk

Covered milk categories

The uploaded amended provision includes:

- Buffalo milk.
- Cow milk.
- Goat milk.
- Sheep milk.
- Camel milk.
- Full-cream milk.
- Toned milk.
- Double-toned milk.
- Skimmed milk.
- Standardized milk.

When fortified, the milk is to undergo pasteurisation, sterilisation, ultra-high-temperature sterilisation or treatment, or boiling, as stated in the provision.

Nutrient	Required range per litre	Permitted source
Vitamin A	270–450 micrograms retinol equivalent	Retinyl acetate or retinyl palmitate
Vitamin D	5–7.5 micrograms	Cholecalciferol or ergocalciferol, only from plant source

International-unit conversions

- Vitamin A: 900–1500 international units per litre.
- Vitamin D: 200–300 international units per litre.

Hypothetical pack-content calculation

For a 500-millilitre pack complying with the milk ranges:

Nutrient	Content range in the pack
Vitamin A	135–225 micrograms retinol equivalent
Vitamin D	2.5–3.75 micrograms

This is multiplication by one-half of the per-litre range, not a new statutory range.

[HPSC EXAM TRAP] The regulation's plant-source restriction applies to the vitamin D source. It does not mean the food vehicle must be plant-based: the provision explicitly covers milk.

8.9. Fortified milk powder

The August 2021 insertion requires:

1. Vitamin A and vitamin D levels such that the final reconstituted milk complies with the fortified-milk table when reconstituted according to label directions.
2. Use only in Government-funded programmes for preparation of reconstituted fortified milk.
3. The following additional label statements.

Required label statements:

- "NOT RECOMMENDED FOR DIRECT CONSUMPTION"
- "ONLY FOR USE UNDER" followed by the name of the Government-funded programme.

- “TO BE CONSUMED ONLY AFTER RECONSTITUTION OF THE ENTIRE CONTENT AS PER THE DIRECTIONS ON THE LABEL.”
- “PACK ONCE OPENED, TO BE CONSUMED ON THE SAME DAY.”

The liquid-milk range must not be copied directly as a per-kilogram dry-powder range.

8.10. Schedule I: atta and maida

Required core nutrients

Nutrient	Range per kilogram	Sources listed for atta and maida
Iron, ordinary listed alternatives	28–42.5 milligrams	Ferrous citrate, ferrous lactate, ferrous sulphate, ferric pyrophosphate, electrolytic iron, ferrous fumarate or ferrous bisglycinate
Iron, sodium feredetate alternative	14–21.25 milligrams	Sodium iron(III) ethylenediaminetetraacetate trihydrate, commonly sodium feredetate
Folic acid	75–125 micrograms	Folic acid
Vitamin B12	0.75–1.25 micrograms	Cyanocobalamin or hydroxocobalamin

The upload notes that the higher ordinary iron range accounts for lower bioavailability.

Source-specific principle

The two iron ranges are alternatives dependent on the listed source. They are not two independent iron additions that must both be made.

The nutrient amount is expressed as iron, not simply as the gross mass of the iron-containing compound.

8.11. Schedule I: rice

The August 2021 amendment replaces the heading **Fortified Raw Rice** with **Fortified Rice**.

Nutrient	Range per kilogram	Sources listed for rice
Iron	28–42.5 milligrams	Ferric pyrophosphate
Iron, alternative	14–21.25 milligrams	Sodium feredetate
Folic acid	75–125 micrograms	Folic acid
Vitamin B12	0.75–1.25 micrograms	Cyanocobalamin or hydroxocobalamin

[HPSC EXAM TRAP] Equal nutrient ranges do not imply equal permitted sources. Ferrous sulphate appears in the atta and maida tables but not in the uploaded rice table.

8.12. Optional micronutrients in atta, maida and rice

Nutrient	Optional range per kilogram	Atta and maida source	Rice source
Zinc	10–15 milligrams	Zinc sulphate	Zinc oxide
Vitamin A	500–750 micrograms retinol equivalent	Retinyl acetate or retinyl palmitate	Retinyl palmitate

Nutrient	Optional range per kilogram	Atta and maida source	Rice source
Thiamine, vitamin B1	1–1.5 milligrams	Thiamine hydrochloride or thiamine mononitrate	Same listed alternatives
Riboflavin, vitamin B2	1.25–1.75 milligrams	Riboflavin or riboflavin 5-prime-phosphate sodium	Same listed alternatives
Niacin, vitamin B3	12.5–20 milligrams	Nicotinamide or nicotinic acid	Same listed alternatives
Pyridoxine, vitamin B6	1.5–2.5 milligrams	Pyridoxine hydrochloride	Same listed source

These additions may be made singly or in combination at the listed levels. Optional does not mean unrestricted.

Multigrain atta

Multigrain atta may be fortified at the same levels as fortified atta **provided it contains more than 50% wheat flour**.

Exactly fifty percent is not more than fifty percent.

8.13. Schedule III: fortified processed foods

Mandatory core nutrients in the listed cereal products and bakery wares

Food group	Examples in the upload	Required nutrient ranges per one hundred grams
Fortified cereal products	Breakfast cereals, pasta and noodles	Iron 1.4–2.7 milligrams; folic acid 8–16 micrograms; vitamin B12 0.08–0.16 microgram
Fortified bakery wares	Bread, biscuits, rusks and buns	Iron 1.4–2.7 milligrams; folic acid 8–16 micrograms; vitamin B12 0.08–0.16 microgram

Sources

For both listed groups, the iron sources include ferrous citrate, ferrous lactate, ferrous sulphate, ferric pyrophosphate, electrolytic iron, ferrous fumarate, ferrous bisglycinate or sodium feredetate.

Vitamin B12 sources are cyanocobalamin or hydroxocobalamin. Folic acid is specified as the nutrient.

Important distinction: Schedule III does not reproduce the separate lower sodium-feredetate iron range used in Schedule I for atta, maida and rice.

Optional additions to these processed-food groups

Nutrient	Range per one hundred grams	Listed source
Zinc	1.0–1.9 milligrams	Zinc sulphate
Vitamin A	48–96 micrograms retinol equivalent	Retinyl acetate or retinyl palmitate
Thiamine	0.1–0.19 milligram	Thiamine hydrochloride or thiamine mononitrate
Riboflavin	0.11–0.22 milligram	Riboflavin or riboflavin 5-prime-phosphate sodium
Niacin	1.3–2.6 milligrams	Nicotinamide or nicotinic acid

Nutrient	Range per one hundred grams	Listed source
Pyridoxine	0.2–0.3 milligram	Pyridoxine hydrochloride

Fortified fruit juices

Nutrient	Range per one hundred millilitres	Source
Vitamin C	6–12 milligrams	Ascorbic acid

Do not confuse the fruit-juice per-one-hundred-millilitre basis with the processed cereal per-one-hundred-gram basis.

8.14. Packaging and labelling under Regulation 7

Packaging

Packaging must consider the nature of the fortificant and its effect on shelf life.

This connects directly to food chemistry: oxygen, light, moisture and contact with other ingredients can influence nutrient retention.

Declaration and logo

Every covered package shall carry:

- The words **“fortified with”**, followed by the name of the fortificant.
- The logo specified in Schedule II.

Exception: The Regulation 7(2) requirements are not required for iodized salt when fortified with iodine.

Tagline

The text states that the package **may also carry**:

“Sampoorna Poshan Swasth Jeevan”

The tagline is therefore optional under this provision.

Graphic specification boundary

Schedule II contains the logo artwork. However, the accessible extracted material does not reliably disclose dimensions, aspect ratios, clear-space measurements or a Pantone 3005 C requirement.

Do not invent those details or confuse an unreadable figure with a verified numerical specification. Exact artwork specifications require the original legible graphic or an authoritative identity guide.

[HPSC EXAM TRAP] Mandatory logo and declaration do not make the accompanying tagline mandatory. Read **shall** and **may** separately.

8.15. Iron-fortified advisory

Under the included August 2021 substitution, every package of food fortified with iron shall carry:

> *People with Thalassemia may take under medical supervision and persons with Sickle Cell Anaemia are advised not to consume iron fortified food products.*

Internal structure of the warning

Group named	Required advisory meaning
People with Thalassemia	May take under medical supervision
Persons with Sickle Cell Anaemia	Advised not to consume iron-fortified food products

The advice is not identical for the two groups. Reversing the clauses changes the declaration.

8.16. Quality assurance under Regulation 5

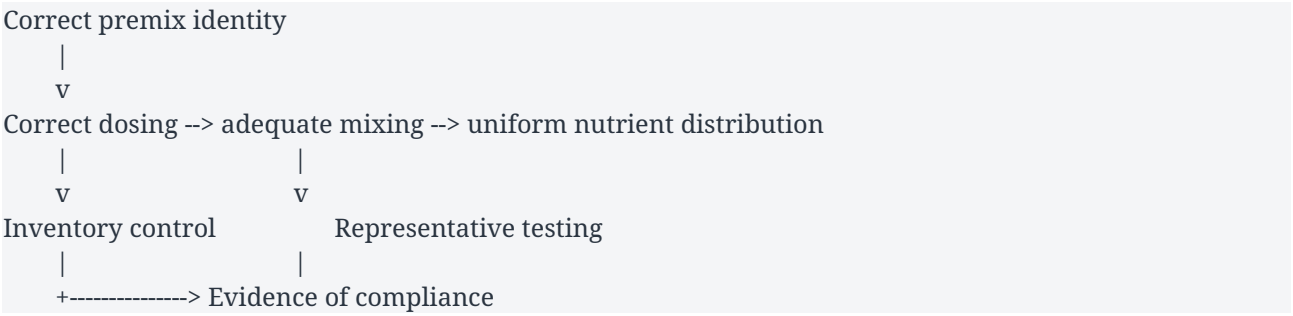
Every manufacturer and packer must give an undertaking on quality assurance and submit evidence to the Food Authority or the designated authority.

The undertaking is required **twice a year** and includes:

1. Certification by a food laboratory notified by the Food Authority regarding compliance.
2. Up-to-date records and continuous inventory of fortificants, including procurement source.
3. Appropriate monitoring at different manufacturing or packing stages.
4. Random testing of fortificants and fortified food.
5. Regular audit of technical equipment and processes.
6. Applicable good manufacturing practices.
7. Reference to generally accepted micronutrient purity criteria, including those recognised by the named pharmacopoeias and international standards bodies.

The named references include the Indian Pharmacopoeia, British Pharmacopoeia, Food Chemical Codex, the Joint Food and Agriculture Organization/World Health Organization Expert Committee on Food Additives, and Codex Alimentarius.

Why this matters chemically



Chemical clarification: One satisfactory sample does not automatically prove uniformity across every pack. Process control and representative sampling are complementary.

8.17. General compliance and promotion

Regulation 6

Voluntary and mandatory fortified foods must comply with generally applicable requirements for manufacture, packing, labelling, handling, distribution and sale. Government-funded distribution is not exempt from compliance.

Regulation 7(5)

Complying manufacturers and packers may make nutrition claims under the referenced framework. This is not permission to make any unqualified health claim.

Regulation 8

The Food Authority is to encourage production, manufacture, distribution, sale and consumption of fortified food, including the stated conventional-breeding or hybridisation approaches, in cooperation with relevant government departments.

Its promotion measures include:

- Advice and promotion for Government-funded food-distribution programmes.
- Public awareness and education.
- Technical assistance, particularly for small manufacturers.
- Capacity for nutrient analysis in notified laboratories and research institutions.

8.18. One-table statutory revision sheet

Vehicle	Core nutrients	Range and basis
Iodized salt	Iodine	15–30 parts per million, dry-weight basis
Double fortified salt	Iodine and iron	Iodine 15–30 parts per million; iron 850–1100 parts per million
Vegetable oil	Vitamins A and D	Vitamin A 6–9.9 micrograms retinol equivalent per gram; vitamin D 0.11–0.16 microgram per gram
Milk	Vitamins A and D	Vitamin A 270–450 micrograms retinol equivalent per litre; vitamin D 5–7.5 micrograms per litre
Atta and maida	Iron, folic acid and vitamin B12	Iron 28–42.5 milligrams, or sodium-feredetate alternative 14–21.25 milligrams, per kilogram; folic acid 75–125 micrograms per kilogram; vitamin B12 0.75–1.25 micrograms per kilogram
Rice	Iron, folic acid and vitamin B12	Same numerical ranges as above, but rice

8.18 Continuation — Revision & Model Answers

SECTION 8.18 — CONTINUATION

Completion of the One-Table Statutory Revision Sheet, Screening Revision Bank and Descriptive Models

Scope and authority: This continuation uses the uploaded *Food Chemistry* textbook and the uploaded *Food Safety and Standards (Fortification of Foods) Regulations, 2018*, Compendium Version III, dated 30 September 2021. Regulatory statements below describe that compendium, including its displayed amendments and compliance dates. They are not a certification of legislation currently in force. The requested B. Srilakshmi textbook and the governing frying-oil notification were not supplied.

Editorial approach: This continuation completes the rice portion of the statutory revision sheet and adds the processed-food standards. It then supplies exactly four count-based questions and complete descriptive model answers. It preserves the existing handout rather than replacing sections whose full text is not visible.

8.18.1. One-Table Statutory Revision Sheet — Remaining Entries

First-principles orientation

Fortification law specifies more than a nutrient name. A complete answer identifies the **food vehicle, permitted chemical source, concentration range, denominator and whether addition is required or optional when the vehicle is fortified**. The denominator is particularly important: Schedule I rice standards are expressed per kilogram, whereas Schedule III cereal and bakery standards are expressed per one hundred grams.

In the table below, iron concentrations mean the amount of iron supplied, not the total mass of its carrier compound. Vitamin A is expressed as retinol equivalent where specified.

Food vehicle and statutory location	Nutrient and legal status within that vehicle	Level	Permitted source in the uploaded compendium
Fortified rice; Schedule I, clause 6	Required iron: first alternative	28–42.5 milligrams per kilogram	Ferric pyrophosphate
Fortified rice; Schedule I, clause 6	Required iron: second alternative	14–21.25 milligrams per kilogram	Sodium iron (III) ethylene diamine tetra acetate trihydrate, also described as sodium feredetate
Fortified rice; Schedule I, clause 6	Required folic acid	75–125 micrograms per kilogram	Folic acid
Fortified rice; Schedule I, clause 6	Required vitamin B12	0.75–1.25 micrograms per kilogram	Cyanocobalamine or hydroxycobalamine, using the spellings in the uploaded table
Fortified rice; Schedule I, clause 6	Optional zinc	10–15 milligrams per kilogram	Zinc oxide
Fortified rice; Schedule I, clause 6	Optional vitamin A	500–750 micrograms retinol equivalent per kilogram	Retinyl palmitate
Fortified rice; Schedule I, clause 6	Optional thiamine, vitamin B1	1–1.5 milligrams per kilogram	Thiamine hydrochloride or thiamine mononitrate
Fortified rice; Schedule I,	Optional riboflavin, vitamin	1.25–1.75 milligrams per	Riboflavin or riboflavin 5′-

Food vehicle and statutory location	Nutrient and legal status within that vehicle	Level	Permitted source in the uploaded compendium
clause 6	B2	kilogram	phosphate sodium
Fortified rice; Schedule I, clause 6	Optional niacin, vitamin B3	12.5–20 milligrams per kilogram	Nicotinamide or nicotinic acid
Fortified rice; Schedule I, clause 6	Optional pyridoxine, vitamin B6	1.5–2.5 milligrams per kilogram	Pyridoxine hydrochloride
Fortified cereal products: breakfast cereals, pasta and noodles; Schedule III, clause 1	Required iron	1.4–2.7 milligrams per 100 grams	Ferrous citrate, ferrous lactate, ferrous sulphate, ferric pyrophosphate, electrolytic iron, ferrous fumarate, ferrous bisglycinate, or sodium iron (III) ethylene diamine tetra acetate trihydrate, also described as sodium feredetate
Fortified cereal products; Schedule III, clause 1	Required folic acid	8–16 micrograms per 100 grams	Folic acid; the table names the nutrient without separately listing another source
Fortified cereal products; Schedule III, clause 1	Required vitamin B12	0.08–0.16 microgram per 100 grams	Cyanocobalamine or hydroxycobalamine
Fortified cereal products; Schedule III, clause 1	Optional zinc	1–1.9 milligrams per 100 grams	Zinc sulphate
Fortified cereal products; Schedule III, clause 1	Optional vitamin A	48–96 micrograms retinol equivalent per 100 grams	Retinyl acetate or retinyl palmitate
Fortified cereal products; Schedule III, clause 1	Optional thiamine, vitamin B1	0.1–0.19 milligram per 100 grams	Thiamine hydrochloride or thiamine mononitrate
Fortified cereal products; Schedule III, clause 1	Optional riboflavin, vitamin B2	0.11–0.22 milligram per 100 grams	Riboflavin or riboflavin 5′-phosphate sodium
Fortified cereal products; Schedule III, clause 1	Optional niacin, vitamin B3	1.3–2.6 milligrams per 100 grams	Nicotinamide or nicotinic acid
Fortified cereal products; Schedule III, clause 1	Optional pyridoxine, vitamin B6	0.2–0.3 milligram per 100 grams	Pyridoxine hydrochloride
Fortified bakery wares: bread, biscuits, rusks and buns; Schedule III, clause 2	Required iron	1.4–2.7 milligrams per 100 grams	Ferrous citrate, ferrous lactate, ferrous sulphate, ferric pyrophosphate, electrolytic iron, ferrous fumarate, ferrous bisglycinate, or sodium iron (III) ethylene diamine tetra acetate trihydrate, also described as sodium feredetate
Fortified bakery wares; Schedule III, clause 2	Required folic acid	8–16 micrograms per 100 grams	Folic acid; the table names the nutrient without separately listing another source
Fortified bakery wares; Schedule III, clause 2	Required vitamin B12	0.08–0.16 microgram per 100 grams	Cyanocobalamine or hydroxycobalamine
Fortified bakery wares;	Optional zinc	1–1.9 milligrams per 100	Zinc sulphate

Food vehicle and statutory location	Nutrient and legal status within that vehicle	Level	Permitted source in the uploaded compendium
Schedule III, clause 2		grams	
Fortified bakery wares; Schedule III, clause 2	Optional vitamin A	48–96 micrograms retinol equivalent per 100 grams	Retinyl acetate or retinyl palmitate
Fortified bakery wares; Schedule III, clause 2	Optional thiamine, vitamin B1	0.1–0.19 milligram per 100 grams	Thiamine hydrochloride or thiamine mononitrate
Fortified bakery wares; Schedule III, clause 2	Optional riboflavin, vitamin B2	0.11–0.22 milligram per 100 grams	Riboflavin or riboflavin 5′-phosphate sodium
Fortified bakery wares; Schedule III, clause 2	Optional niacin, vitamin B3	1.3–2.6 milligrams per 100 grams	Nicotinamide or nicotinic acid
Fortified bakery wares; Schedule III, clause 2	Optional pyridoxine, vitamin B6	0.2–0.3 milligram per 100 grams	Pyridoxine hydrochloride
Fortified fruit juices; Schedule III, clause 3	Required vitamin C when fortified under this category	6–12 milligrams per 100 millilitres	Ascorbic acid

Essential qualifications to the completed sheet

1. **The rice iron alternatives are alternatives, not cumulative additions.** The higher ferric-pyrophosphate range is explained in the schedule as accounting for lower bioavailability. Do not add both prescribed iron ranges together.
2. **Rice has a narrower iron-source list than atta and maida.** Their additional permitted ferrous salts and electrolytic iron must not automatically be transferred to rice.
3. **Optional rice sources differ from flour sources.** Rice specifies zinc oxide and retinyl palmitate. Atta and maida specify zinc sulphate and permit retinyl acetate or retinyl palmitate.
4. **Optional micronutrients may be added singly or in combination**, but an elected addition must comply with its applicable source and range.
5. **"When fortified, shall contain" does not mean all products in that category must universally undergo fortification.** Regulation 3 separately addresses how mandatory fortification may be specified.
6. **The August 2021 amendment changes the heading "Fortified Raw Rice" to "Fortified Rice".** It was stated to be in force from 27 August 2021, with compliance by 1 March 2022.
7. **Schedule III is not Schedule I scaled down mechanically.** Use the processed-food table itself.
8. **High Fat Sugar Salt foods are excluded from the Fortified Processed Foods category.** Listing biscuits under bakery wares does not override this exclusion for a biscuit that meets the applicable definition.
9. **Regulation 3(6)** provides that fortified processed food shall provide 15–30% of the Indian adult recommended dietary allowance of micronutrients, based on an average intake of 600 kilocalories from processed foods, approximately one-third of the daily adult energy requirement used in that provision. This framework does not replace the specific Schedule III concentration tables.
10. **Regulation 4(2)** requires micronutrient levels in fortified processed food to fall within Schedule III ranges. The Schedule I label-tolerance amendment should not be used as an assumed universal extension of every Schedule III limit.

[HPSC EXAM TRAP] "All permitted iron salts are interchangeable in every fortified cereal food" is false. Identify the specific vehicle before selecting a source.

Unit-conversion verification

These calculations compare denominators; they do not create substitute legal standards.

Iron in processed cereal and bakery products:

$$1\ kilogram = 10\times100\ grams$$

$$1.4\times10 = 14\ milligrams\ per\ kilogram$$

$$2.7\times10 = 27\ milligrams\ per\ kilogram$$

Thus, 1.4–2.7 milligrams per 100 grams is numerically equivalent to **14–27 milligrams per kilogram**. This is not identical to either of the Schedule I rice iron ranges.

Folic acid in processed cereal and bakery products:

$$8\times10 = 80\ micrograms\ per\ kilogram$$

$$16\times10 = 160\ micrograms\ per\ kilogram$$

The converted range is **80–160 micrograms per kilogram**, not the Schedule I staple range of 75–125 micrograms per kilogram.

Vitamin B12 in processed cereal and bakery products:

$$0.08\times10 = 0.8\ microgram\ per\ kilogram$$

$$0.16\times10 = 1.6\ micrograms\ per\ kilogram$$

The converted range is **0.8–1.6 micrograms per kilogram**, not the Schedule I staple range of 0.75–1.25 micrograms per kilogram.

Vitamin C in fruit juice:

$$1\ litre = 10\times100\ millilitres$$

$$6\times10 = 60\ milligrams\ per\ litre$$

$$12\times10 = 120\ milligrams\ per\ litre$$

The converted range is **60–120 milligrams per litre**. Do not convert millilitres to grams without the necessary density information.

8.19. STAGE 1 REVISION BANK

Exactly Four Count-Based Pair Questions

Instructions: Each question contains three pairs. Judge the correctness of the entire pair, including its qualifier. Assign 1 to a correct pair and 0 to an incorrect pair, then calculate:

$$N = c_1+c_2+c_3$$

The options are identical for every question:

- **A. Only 1 pair**
- **B. Only 2 pairs**
- **C. All 3 pairs**
- **D. None**

The answer explanations use the uploaded textbook and compendium, with explicitly identified scientific clarification where necessary. No question depends on an unverified current-law frying threshold or colour limit.

Question 1 — Lipids and Repeated Frying

How many of the following pairs are correctly matched?

Pair	Subject	Matched statement
1	Milk fat	Contains appreciable short-chain fatty acids, with butyric acid as a characteristic constituent

Pair	Subject	Matched statement
2	Polymerisation during deep-fat frying	Produces coupled lipid products and contributes to increasing oil viscosity
3	Peroxide content during repeated frying	Must increase continuously because hydroperoxides cannot decompose at frying temperature

Options: A. Only 1 pair; B. Only 2 pairs; C. All 3 pairs; D. None.

Step-by-step verification

Pair 1: Correct. Lesson 8 identifies appreciable short-chain fatty acids in milk fat and describes butyric acid as characteristic. This does not claim that butyric acid is the most abundant milk-fat fatty acid.

$$c_1=1$$

Pair 2: Correct. Lesson 9 connects dimeric and polymeric products with increased viscosity. Radical coupling and thermal alteration create larger lipid-derived structures.

$$c_2=1$$

Pair 3: Incorrect. Lesson 9 describes peroxide content rising and subsequently falling. Hydroperoxides can decompose at frying temperature. The absolute word “must” and the claimed inability to decompose make the pair false.

$$c_3=0$$

Arithmetic:

$$N=c_1+c_2+c_3=1+1+0=2$$

Answer: B. Only 2 pairs.

[HPSC EXAM TRAP] A decrease in peroxide content during continued heating can accompany worsening oil quality, not recovery.

Question 2 — Enzyme Inhibition

How many of the following pairs are correctly matched under the elementary Michaelis–Menten inhibition models?

Pair	Inhibition type	Matched kinetic change
1	Competitive inhibition	Apparent Michaelis constant increases while maximum reaction velocity remains unchanged
2	Pure noncompetitive inhibition	Apparent Michaelis constant remains unchanged while maximum reaction velocity decreases
3	Uncompetitive inhibition	Both apparent Michaelis constant and maximum reaction velocity decrease

Options: A. Only 1 pair; B. Only 2 pairs; C. All 3 pairs; D. None.

Step-by-step verification

Pair 1: Correct. Competitive binding increases the substrate concentration needed to reach a given fraction of maximum velocity. Sufficient substrate can overcome the inhibition in this model, so maximum velocity is unchanged.

$$c_1=1$$

Pair 2: Correct. The word **pure** is essential. Equal inhibitor affinity for free enzyme and enzyme–substrate complex gives an unchanged apparent Michaelis constant and a reduced maximum velocity. Unequal affinities produce mixed inhibition instead.

$$c_2=1$$

Pair 3: Correct. An uncompetitive inhibitor binds the enzyme–substrate complex. In the elementary model, both parameters decrease by the same factor, giving parallel Lineweaver–Burk lines.

$$c_3=1$$

Arithmetic:

$$N=c_1+c_2+c_3=1+1+1=3$$

Answer: C. All 3 pairs.

Scientific caveat: These parameter changes describe the stated ideal models, not every multisubstrate, allosteric or irreversible inhibition mechanism.

[HPSC EXAM TRAP] Binding away from the active site does not by itself establish pure noncompetitive kinetics.

Question 3 — Pigments and Synthetic Colours

How many of the following pairs are correctly matched according to the uploaded textbook?

Pair	Pigment or colour	Matched classification
1	Beta-carotene	Hydrocarbon carotene rather than an oxygenated xanthophyll
2	Anthocyanins	Water-insoluble pigments restricted to the lipid phase
3	Fast Green FCF	Azo dye in the textbook's permitted synthetic-colour table

Options: A. Only 1 pair; B. Only 2 pairs; C. All 3 pairs; D. None.

Step-by-step verification

Pair 1: Correct. The textbook distinguishes hydrocarbon carotenes from oxygenated xanthophylls. Beta-carotene is a hydrocarbon carotene.

$$c_1=1$$

Pair 2: Incorrect. Lesson 20 describes anthocyanins as water-soluble pigments associated with red-to-blue colours in numerous fruits and vegetables. The proposed restriction to the lipid phase is false.

$$c_2=0$$

Pair 3: Incorrect. The textbook lists Fast Green FCF as a **triarylmethane** colour, not an azo dye.

$$c_3=0$$

Arithmetic:

$$N=c_1+c_2+c_3=1+0+0=1$$

Answer: A. Only 1 pair.

Legal caveat: Identification in the textbook's colour table does not establish permission in every food category or independently verify the currently applicable maximum level.

[HPSC EXAM TRAP] Colour shade, chemical class and legal food-category permission are three separate facts.

Question 4 — Fortification Regulations

How many of the following pairs are correctly matched according to the uploaded 30 September 2021 compendium?

Pair	Statutory subject	Matched requirement
1	Fortification tagline	“Sampoorna Poshan Swasth Jeevan” must appear on every fortified-food package under Regulation 7(2)
2	Iodized salt fortified with iodine	Regulation 7(2) gives no exception from its fortification-declaration and prescribed-logo requirements
3	Fortified rice iron sources	All seven non-sodium-feredetate iron sources listed for atta are also listed for rice

Options: A. Only 1 pair; B. Only 2 pairs; C. All 3 pairs; D. None.

Step-by-step verification

Pair 1: Incorrect. Regulation 7(2) states that the package **may also carry** the tagline. The tagline is optional under that wording, not mandatory.

$$c_1=0$$

Pair 2: Incorrect. The proviso expressly exempts iodized salt fortified with iodine from the requirements of Regulation 7(2). This is not an exemption from all applicable food-labelling law.

$$c_2=0$$

Pair 3: Incorrect. Rice lists ferric pyrophosphate or sodium feredetate. Its source list is not identical to the broader atta list.

$$c_3=0$$

Arithmetic:

$$N=c_1+c_2+c_3=0+0+0=0$$

Answer: D. None.

[HPSC EXAM TRAP] Distinguish “shall”, “may”, “when fortified” and an express proviso. Each can change the answer even when the nutrient names are familiar.

Revision-bank score verification

The following arithmetic assumes **one mark per correct answer, a deduction of 0.25 mark per incorrect answer and no deduction for an unanswered question**, matching the preparation pattern supplied by the user rather than an independently verified recruitment notice.

Let **C** be correct answers, **W** incorrect answers and **U** unanswered questions.

$$C+W+U=4$$

$$Score=C-0.25W$$

Performance	Calculation	Net score
Four correct	$4 - 0.25 \times 0$	4.00
Three correct and one incorrect	$3 - 0.25 \times 1$	2.75
Two correct and two incorrect	$2 - 0.25 \times 2$	1.50
One correct and three incorrect	$1 - 0.25 \times 3$	0.25
Four incorrect	$0 - 0.25 \times 4$	−1.00
Three correct and one unanswered	$3 - 0.25 \times 0$	3.00

Answer key: Question 1 — B; Question 2 — C; Question 3 — A; Question 4 — D.

8.20. STAGE 2 DESCRIPTIVE MODEL ANSWER — 10 MARKS

Question

“Critically evaluate the physical, chemical, and analytical alterations occurring in edible oil during repeated deep-fat frying. Discuss why Peroxide Value alone is inadequate for assessing used frying oil quality.”

Suggested marking breakdown

This is a teaching allocation, not an official HPSC marking scheme.

Component	Suggested marks
Introduction: coupled heat transfer, moisture transfer and lipid reaction system	1.0
Physical alterations and their interpretation	1.5
Hydrolysis, oxidation and polymerisation mechanisms	3.0
Analytical alterations and selection of complementary measures	2.0
Critical evaluation of peroxide value	2.0
Integrated conclusion	0.5
Total	**10.0**

Complete model answer

Introduction

Deep-fat frying exposes edible oil to high temperature, commonly around 180 degrees Celsius in the uploaded textbook, while food continuously supplies water, solids and sometimes endogenous lipids. Repeated heating and cooling therefore produce a changing reaction mixture rather than merely warming an unchanged oil. Hydrolysis, thermal oxidation and polymerisation progressively alter physical behaviour, composition and analytical indices.

1. Physical alterations

Moisture release and steam stripping: Water leaving the food becomes steam, agitates the oil and strips some volatile degradation products. Consequently, volatile-product concentrations may approach a plateau even while nonvolatile degradation continues. A steam blanket may reduce oxygen availability above the surface, but it does not eliminate oxidation.

Oil uptake and lipid exchange: The textbook reports approximately 5–40% fat by mass in finished fried foods, with potato chips illustrated at approximately 35%. These are product-dependent examples, not universal absorption limits. Fat released from foods such as chicken can change the composition and oxidative stability of the frying medium.

Increasing viscosity: Coupled and polymerised lipid products increase molecular size and viscosity. Oil may drain less effectively and the product may become greasy.

Foaming, darkening and deposits: Polar degradation products and accumulated food material contribute to persistent foaming, darkening, gums and residues. Normal steam bubbling during frying must be distinguished from deterioration-associated persistent foam.

Reduced smoke point: Accumulation of free fatty acids contributes to smoking at lower temperatures. Smoke point is nevertheless influenced by other oil constituents and cannot alone establish compliance.

2. Chemical alterations

A. Hydrolysis

Food-derived water promotes cleavage of triacylglycerol ester bonds under frying conditions:

```
Triacylglycerol + water --> diacylglycerol + free fatty acid
Diacylglycerol + water --> monoacylglycerol + free fatty acid
Monoacylglycerol + water --> glycerol + free fatty acid
```

Complete hydrolysis consumes three water molecules per triacylglycerol. The resulting free fatty acids increase acidity and contribute to a lower smoke point. This high-temperature hydrolysis does not require the original food lipase to remain active.

B. Thermal oxidation

Unsaturated lipid chains undergo oxygen-associated oxidation. A simplified chain mechanism is:

```
Lipid radical + oxygen --> lipid peroxyl radical
Lipid peroxyl radical + lipid --> hydroperoxide + new lipid radical
Hydroperoxide --> reactive intermediates --> secondary products
```

Hydroperoxide decomposition generates aldehydes, ketones, alcohols, acids and hydrocarbons. Volatile products contribute to flavour changes and may partly escape with steam. Oxygenated nonvolatile compounds remain in the oil. Temperature, oxygen exposure, metals, oil composition and heating duration govern the extent of deterioration.

C. Polymerisation and thermal alteration

Reactive lipid chains combine through carbon–carbon and oxygen-containing linkages. Cyclic monomers, dimers, trimers and higher polymers can form. Their accumulation increases viscosity, foaming, gum deposition and potentially oil retention by food. The textbook associates greater unsaturation and longer frying time with increased polymerisation under comparable conditions.

3. Analytical alterations

Measurement	Typical change or relevance	Essential qualification
Acid value or free fatty acid content	Usually increases as free acids accumulate	Indicates free acidity, not the entire degradation burden

Measurement	Typical change or relevance	Essential qualification
Iodine value	Commonly decreases as unsaturation is altered or consumed	Does not directly quantify every harmful product
Peroxide value	May increase initially and then decrease	Reflects a transient primary-oxidation pool
Viscosity	Increases with polymerised-product accumulation	Measurement temperature must be controlled
Refractive index	Changes; an increase is reported in the textbook	Requires consistent temperature and method
Saponification value	An increase is reported for the mixed degradation system in the textbook	Competing cleavage and coupling processes prevent a universal prediction from polymerisation alone
Total Polar Compounds	Broader assessment of polar constituents and deterioration products	Method-defined fraction; not synonymous with peroxide value
Secondary-oxidation or carbonyl measurements	Can complement primary-oxidation assessment	Method suitability and food-derived interference must be considered

A rise in saponification value does not contradict polymer formation automatically: actual used oil contains a mixture of polymers, cleavage products, partial glycerides and free acids. The ideal inverse relationship between saponification value and pure triacylglycerol molar mass cannot be applied mechanically to that mixture.

4. Why peroxide value alone is inadequate

Peroxide value measures peroxide-related primary oxidation, conventionally expressed as milliequivalents of active oxygen per kilogram. It does not measure cumulative deterioration.

The central kinetic distinction is:

$$\frac{d[Hydroperoxides]}{dt}=Formation\ rate-Decomposition\ rate$$

If decomposition exceeds formation, hydroperoxide concentration falls despite continuing deterioration. High frying temperatures favour rapid decomposition, and the textbook explicitly describes peroxide content rising and later falling.

Peroxide value alone is therefore inadequate for five reasons:

1. **Hydroperoxides are unstable at frying temperatures.** A small measured pool may reflect rapid destruction rather than little oxidation.
2. **Secondary oxidation products are not comprehensively measured.** Aldehydes, ketones and other products may accumulate after hydroperoxides disappear.
3. **Hydrolysis is a different deterioration pathway.** Free fatty acids can increase without a corresponding peroxide-value increase.
4. **Polymerisation is not represented adequately.** High-molecular-mass deterioration products can persist when peroxide value is low.
5. **Thermal and sampling history matter.** Cooling and oxygen exposure can alter the measured primary-oxidation pool. Oils with different histories cannot be ranked reliably by this index alone.

```
Fresh oil ----------> low peroxide value
Severely heated oil with decomposition -> potentially low peroxide value

Same index range does not imply the same quality history.
```

5. Integrated assessment and control

Used frying oil should be assessed through a combination of processing history, acidity, appropriate primary- and secondary-oxidation measures, polar-product or polymer assessment, and physical observations. Good

practice includes controlling temperature, avoiding prolonged idle heating, managing food debris and preventing uncontrolled repeated thermal cycling.

Continuous replenishment can limit accumulation under some operating conditions, as explained by the textbook, but adding fresh oil does not prove that the resulting mixture is compliant or justify indefinite use.

Conclusion

Repeated frying produces interconnected hydrolytic, oxidative and polymerisation changes. Peroxide value is useful for primary oxidation, but its rise-and-fall behaviour makes it unreliable as a standalone used-oil quality test. A defensible decision combines complementary analytical measures with the verified applicable regulatory criterion.

Worked analytical illustration

This is a hypothetical calculation, not measured data from the uploads.

For iodometric peroxide-value measurement:

$$Peroxide\ value=\frac{1000(V_s-V_b)N_t}{m}$$

Where:

- **Vs** is sample sodium-thiosulphate titre in millilitres.
- **Vb** is blank titre in millilitres.
- **Nt** is sodium-thiosulphate normality.
- **m** is oil sample mass in grams.

Assume sample titre 1.50 millilitres, blank titre 0.10 millilitre, normality 0.01 and sample mass 5.00 grams.

$$V_s-V_b=1.50-0.10=1.40\ millilitres$$

$$1000\times1.40\times0.01=14$$

$$Peroxide\ value=14\div5.00=2.8\ milliequivalents\ per\ kilogram$$

Permissible inference: The assay detects the stated peroxide-related oxidation level under its conditions.

Impermissible inference: The oil must be fresh, safe or legally suitable for further frying. Those conclusions require additional evidence.

Legal caveat: Total Polar Compounds thresholds

The requested revision targets are **15% for fresh oil and 25% for used frying oil**. Neither uploaded document establishes their governing legal provision or exact boundary wording. Accordingly, this source-limited model does not represent them as verified statutory limits.

Before quoting these figures as law, verify the relevant food-products standard or notification, its current amendment status, the relevant oil category, the method and the exact discard rule. “Not more than 25%” and “25% or more shall not be used” have different consequences at exactly 25%. Do not silently equate them.

[HPSC EXAM TRAP] “Low peroxide value proves that a used frying oil is suitable for reuse” is scientifically invalid even before the legal criterion is considered.

Examination writing skeleton

1. Open with the coupled heat–moisture–reaction system.
2. State viscosity, foaming, darkening and smoke-point changes.
3. Explain hydrolysis, oxidation and polymerisation with a compact pathway.
4. Compare acid, iodine, peroxide and broader deterioration measures.
5. Give the formation-minus-decomposition explanation of peroxide-value inadequacy.
6. Close with integrated assessment and the applicable verified legal criterion.

8.21. STAGE 2 DESCRIPTIVE MODEL ANSWER — 15 MARKS

Question

“Examine the statutory framework of Food Fortification under FSSAI Regulations. Detail the standards for fortified staple foods (Schedule I), mandatory label advisories, and the quality assurance obligations imposed on manufacturers.”

Suggested marking breakdown

This is a teaching allocation, not an official HPSC marking scheme.

Component	Suggested marks
Definition, scope and general principles	2.0
Schedule I staple standards, permitted sources and units	6.0
Packaging, declarations, logo, exception and iron advisory	3.0
Quality assurance and general compliance obligations	2.0
Label tolerance, amendment distinctions and critical evaluation	1.5
Conclusion	0.5
Total	**15.0**

Complete model answer

Introduction

The Food Safety and Standards (Fortification of Foods) Regulations, 2018 establish a food-vehicle-specific framework for deliberately increasing essential micronutrients to improve nutritional quality and provide public-health benefit with minimal health risk. The uploaded 30 September 2021 compendium connects nutrient standards with permitted sources, packaging, labelling, quality assurance and generally applicable food law.

Fortification is therefore not simply a nutrient-addition claim: it is a regulated process requiring suitable composition, reliable manufacturing control and appropriate consumer information.

1. Definitions and statutory scope

Under Regulation 2:

- **Fortification** means deliberately increasing the content of essential micronutrients in a food to improve its nutritional quality and provide public-health benefit with minimal health risk.
- **Fortificant** means a substance added to food to provide micronutrients; the definition excludes nutraceuticals and foods for Special Dietary Uses.
- **Fortified food** is a food specified under the Food Products Standards and Food Additives Regulations, 2011 that has undergone fortification in accordance with these regulations.
- **Micronutrients** include essential dietary vitamins, minerals and trace elements needed in very small quantities and important to development, disease prevention and wellbeing.
- **Staple foods** are articles intended for daily mass consumption, including rice, wheat, wheat flour, atta, maida, oil, salt, milk and other designated staples.
- **Quality assurance** means systematic measures throughout manufacturing or packing to ensure compliance of the finished fortified food.

The regulations state that they supersede earlier fortification standards under the Act, while preserving the specified exception concerning regulations on nutraceuticals and foods for Special Dietary Uses.

2. General principles under Regulation 3

Micronutrient addition may contribute to preventing or correcting demonstrated deficiency, improving inadequate nutritional status, meeting recommended intake, maintaining or improving health, and maintaining or improving nutritional quality.

Mandatory fortification must be grounded in the severity and extent of public-health need, demonstrated by generally accepted scientific evidence. The Food Authority may specify mandatory fortification of a staple on direction of the Government of India. Hence, **"when fortified, shall" is not equivalent to universal compulsory fortification of every product**.

Where iron is used as a nutrient source, **heme iron shall not be used in any form in any article of food** under Regulation 3(4).

For fortified processed foods, Regulations 3(5)–3(7) provide for preparation using fortified food articles that may be cereals or milk, a 15–30% Indian adult recommended-dietary-allowance framework based on 600 kilocalories from processed foods, and exclusion of High Fat Sugar Salt foods from that category.

3. Schedule I standards for fortified staples

A. Iodized salt and double fortified salt

Product or component	Standard	Source
Iodine in iodized salt and double fortified salt, using the August 2021 substituted provision	15–30 parts per million on a dry-weight basis	Potassium iodate
Iron in double fortified salt	850–1100 parts per million as iron	Ferrous sulphate or ferrous fumarate

The earlier displayed iodine provision distinguished 20–30 parts per million at manufacture and 15–30 at distribution, including retail. The August 2021 amendment substitutes the single 15–30 parts-per-million dry-weight range. The compendium states commencement of that amendment on 27 August 2021 and compliance by 1 March 2022.

The salt footnotes also matter:

- Where an anticaking agent is added, total water-insoluble matter shall not exceed 2.2%.
- Sodium chloride content in the applicable iodized-salt footnote shall not be less than 97% by weight on a dry basis; other referenced salt parameters also apply.
- Double fortified salt may contain food-grade hydroxypropyl methyl cellulose, titanium dioxide, fully hydrogenated soybean oil and sodium hexametaphosphate.
- The double-fortified-salt footnote permits an anticaking agent not exceeding 2% on a dry-weight basis and specifies the 2.2% water-insoluble-matter ceiling where such an agent is used.

The express double-fortified-salt provision is why a general textbook statement about inorganic colouring matter must not be misread as prohibiting every inorganic additive in every food.

B. Fortified vegetable oil

Nutrient	Level per gram of oil	Source
Vitamin A	6–9.9 micrograms retinol equivalent	Retinyl acetate or retinyl palmitate
Vitamin D	0.11–0.16 microgram	Cholecalciferol or ergocalciferol, only from plant source

Both listed vitamins are specified when vegetable oil is fortified under this category. The plant-source qualification concerns vitamin D; it does not mean that only vitamin D2 can qualify.

C. Fortified milk

The category includes species-identified buffalo, cow, goat, sheep and camel milk, as well as full cream, toned, double toned, skimmed and standardized milk. The milk is to undergo pasteurization, sterilization, ultra-high-temperature sterilization or treatment, or boiling.

Nutrient	Level per litre	Source
Vitamin A	270–450 micrograms retinol equivalent	Retinyl acetate or retinyl palmitate
Vitamin D	5–7.5 micrograms	Cholecalciferol or ergocalciferol, only from plant source

These are per-litre milk standards, not per-gram oil standards.

D. Fortified atta and maida

Both vehicles require added iron, folic acid and vitamin B12 when fortified under their respective Schedule I provisions.

Nutrient	Level per kilogram	Permitted source
Iron: first alternative	28–42.5 milligrams	Ferrous citrate, ferrous lactate, ferrous sulphate, ferric pyrophosphate, electrolytic iron, ferrous fumarate or ferrous bisglycinate
Iron: second alternative	14–21.25 milligrams	Sodium iron (III) ethylene diamine tetra acetate trihydrate, also described as sodium feredetate
Folic acid	75–125 micrograms	Folic acid
Vitamin B12	0.75–1.25 micrograms	Cyanocobalamine or hydroxycobalamine

The schedule explains the higher level of the first iron alternative as accounting for lower bioavailability.

Multigrain atta may use the fortified-atta levels provided it contains **more than 50% wheat flour**. Exactly 50% does not satisfy the literal “more than” condition.

E. Fortified rice

Nutrient	Level per kilogram	Permitted source
Iron: first alternative	28–42.5 milligrams	Ferric pyrophosphate
Iron: second alternative	14–21.25 milligrams	Sodium iron (III) ethylene diamine tetra acetate trihydrate, also described as sodium feredetate
Folic acid	75–125 micrograms	Folic acid
Vitamin B12	0.75–1.25 micrograms	Cyanocobalamine or hydroxycobalamine

Rice and flour share the principal nutrient ranges, but **not the complete iron-source list**. The August 2021 amendment replaces the heading “Fortified Raw Rice” with “Fortified Rice”.

F. Optional additions to atta, maida and rice

Optional nutrient	Level per kilogram	Atta and maida source	Rice source
Zinc	10–15 milligrams	Zinc sulphate	Zinc oxide
Vitamin A	500–750 micrograms retinol equivalent	Retinyl acetate or retinyl palmitate	Retinyl palmitate
Thiamine, vitamin B1	1–1.5 milligrams	Thiamine hydrochloride or thiamine mononitrate	Thiamine hydrochloride or thiamine mononitrate
Riboflavin, vitamin B2	1.25–1.75 milligrams	Riboflavin or riboflavin 5′-phosphate sodium	Riboflavin or riboflavin 5′-phosphate sodium
Niacin, vitamin B3	12.5–20 milligrams	Nicotinamide or nicotinic acid	Nicotinamide or nicotinic acid
Pyridoxine, vitamin B6	1.5–2.5 milligrams	Pyridoxine hydrochloride	Pyridoxine hydrochloride

These may be added singly or in combination. Optional status does not make their specified source and level irrelevant when an addition is made.

G. Fortified milk powder

The August 2021 insertion requires vitamin A and vitamin D levels that yield compliant fortified milk after reconstitution according to the label. The powder is restricted to Government-funded programmes for preparing reconstituted fortified milk.

Its label must additionally state that:

1. It is not recommended for direct consumption.
2. It is only for use under the named Government-funded programme.
3. It is to be consumed only after reconstitution of the entire contents according to the label directions.
4. Once opened, the pack is to be consumed on the same day.

The compendium states that this amendment was in force from 27 August 2021, with compliance by 1 March 2022.

4. Packaging, declaration, logo and mandatory iron advisory

Packaging

Regulation 7(1) requires packaging to consider the nature of the added fortificant and its effect on shelf life. Appropriate packaging must therefore protect nutrient retention and the food's quality; merely adding a nutrient at manufacture does not remove storage-related obligations.

Fortification declaration and logo

Regulation 7(2) requires the package to carry the words **"fortified with" followed by the name of the fortificant**, together with the prescribed Schedule II logo.

The same provision says the package **may also carry** the tagline **"Sampoorna Poshan Swasth Jeevan"** under the logo. The tagline is optional under this wording.

Express exception: The requirements of Regulation 7(2) do not apply to iodized salt when fortified with iodine. This does not exempt that salt from every other labelling requirement, nor does it automatically exempt iron-containing double fortified salt.

The accessible Schedule II extraction does not establish numerical graphic dimensions or a Pantone specification. These must not be invented or attributed to the uploaded text.

Iron-fortified food advisory

The substituted Regulation 7(4) requires every package of iron-fortified food to carry:

> *People with Thalassemia may take under medical supervision and persons with Sickle Cell Anaemia are advised not to consume iron fortified food products.*

This wording distinguishes medical supervision for people with thalassemia from the non-consumption advice for persons with sickle cell anaemia. Do not replace the two-part advisory with the earlier thalassemia-only statement.

The compendium states that this amendment was in force from **27 August 2021**, with compliance by **1 March 2022**.

Regulation 7 also applies the general packaging and labelling provisions named in the compendium and permits compliant manufacturers and packers to make nutrition claims under the referenced framework. This is not authorization for unsupported therapeutic or disease-cure claims.

5. Quality assurance obligations under Regulation 5

Every manufacturer and packer must give an undertaking on quality assurance and submit evidence to the Food Authority or its designated authority. The undertaking is required **twice a year** and must include:

1. **Notified-laboratory certification:** Certification by a food laboratory notified by the Food Authority that the fortified food complies with the Act, regulations and standards.
2. **Records and inventory:** Up-to-date records and continuous inventory of fortificants, including their procurement sources.
3. **Stage-wise monitoring:** Appropriate monitoring procedures at different manufacturing or packing stages.
4. **Random testing:** Testing of fortificants and fortified food.
5. **Equipment and process audit:** Regular audit of technical equipment and processes.
6. **Good manufacturing practices:** Application of practices specified by the Food Authority from time to time.
7. **Purity criteria:** The provision allows adoption of generally accepted micronutrient purity criteria associated with the Indian Pharmacopoeia, British Pharmacopoeia, Food Chemical Codex, Joint Food and Agriculture Organization or World Health Organization Expert Committee on Food Additives, or Codex Alimentarius.

The permissive wording concerning adoption of referenced purity criteria should not be transformed into a requirement to use all listed references simultaneously.

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Permitted source and supplier verification
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                v
Fortificant inventory and identity control
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                v
Controlled addition and manufacturing-stage monitoring
                |
                v
Random testing and notified-laboratory certification
                |
                v
Packaging, label verification and compliant distribution
                |
                v
Records, regular audits and twice-yearly undertaking
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Regulation 6 requires generally applicable compliance for both voluntarily and mandatorily fortified food. Manufacturing, packing, labelling, handling, distribution and sale remain subject to the Act and regulations, whether undertaken for profit or through a Government-funded programme.

6. Label range and analytical tolerance

The September 2021 substitution of Regulation 4(1) requires the **added micronutrients declared on the label** to fall within Schedule I ranges. It allows a tolerance of **minus 10% of the declared total value on the label** for compliance and analysis at any point in time, while expressly requiring compliance with the Schedule I maximum for added micronutrients.

The compendium states that this amendment was in force from **22 September 2021**, with compliance by **1 April 2022**.

Three distinctions follow:

- Declared added micronutrient levels must be within the applicable prescribed range.
- A negative analytical tolerance is not an authorization for a ten-percent upward excess.
- “Added micronutrients” and “declared total value” are different phrases in the provision and must not be silently treated as identical where intrinsic nutrients contribute.

7. Critical evaluation

The framework combines public-health purpose with product-specific limits and consumer safeguards. Permitted-source restrictions address more than nutrient quantity: bioavailability, food compatibility and stability affect the suitability of the fortificant. The different iron-source ranges illustrate the importance of chemical form.

Effective implementation nevertheless depends on manufacturing uniformity, appropriate mixing, representative sampling, nutrient retention, label accuracy and documented quality assurance. A compliant-looking logo is not a substitute for actual nutrient content or reliable process control.

Fortified processed foods require separate Schedule III assessment, and High Fat Sugar Salt exclusion prevents nutrient addition from automatically converting a disqualified processed food into an eligible fortified processed product.

Conclusion

Fortification under the uploaded FSSAI framework is a coordinated system of public-health justification, vehicle-specific nutrient standards, permitted sources, packaging and labelling safeguards, and documented quality assurance. Strong compliance requires both accurate nutrient addition and continued verification of the food placed on the market.

Worked statutory calculations

Calculation 1 — Parts per million in double fortified salt

For mass concentration:

$$1\ part\ per\ million=1\ milligram\ per\ kilogram$$

Therefore, the prescribed iron range of 850–1100 parts per million as iron corresponds to:

$$850–1100\ milligrams\ of\ iron\ per\ kilogram\ of\ salt$$

For a hypothetical 100-kilogram quantity assessed on the applicable mass basis:

$$850\times100=85000\ milligrams=85\ grams$$

$$1100\times100=110000\ milligrams=110\ grams$$

The corresponding iron amount is **85–110 grams**. It is not 85–110 grams of ferrous sulphate or ferrous fumarate: carrier mass depends on iron fraction, chemical form and purity.

Calculation 2 — Oil vitamin units

The schedule states:

$$1\ international\ unit\ of\ vitamin\ A=0.3\ microgram\ retinol\ equivalent$$

Thus:

$$6\div0.3=20\ international\ units\ per\ gram$$

$$9.9\div0.3=33\ international\ units\ per\ gram$$

Oil vitamin A range: **20–33 international units per gram**.

For vitamin D:

$$1\ international\ unit=0.025\ microgram$$

$$0.11\div0.025=4.4\ international\ units\ per\ gram$$

$$0.16\div0.025=6.4\ international\ units\ per\ gram$$

Oil vitamin D range: **4.4–6.4 international units per gram**. These conversions do not remove the plant-source requirement.

Calculation 3 — Milk vitamin units

Vitamin A:

$$270\div0.3=900\ international\ units\ per\ litre$$

$$450\div0.3=1500\ international\ units\ per\ litre$$

Vitamin D:

$$5\div0.025=200\ international\ units\ per\ litre$$

$$7.5\div0.025=300\ international\ units\ per\ litre$$

Milk ranges are therefore **900–1500 international units per litre for vitamin A** and **200–300 international units per litre for vitamin D**.

Calculation 4 — Negative label tolerance

Assume, solely for illustration, that the declared total value of a nutrient is 35 milligrams per kilogram and that the relevant added-nutrient declaration is within its applicable Schedule I range.

$$10\%\ of\ 35=0.10\times35=3.5\ milligrams\ per\ kilogram$$

$$35-3.5=31.5\ milligrams\ per\ kilogram$$

The numerical minus-ten-percent point is **31.5 milligrams per kilogram**.

This calculation illustrates the stated tolerance only. It does not authorize a declaration outside the prescribed range, abolish the maximum added-nutrient limit, establish the treatment of every intrinsic-nutrient contribution, or create a universal Schedule III tolerance.

Calculation 5 — Twice-yearly quality assurance undertaking

$$1\ year\times2\ undertakings\ per\ year=2\ undertakings$$

“Twice a year” is the stated submission frequency. It does not mean that monitoring, inventory maintenance or testing occurs only twice a year. The listed controls include continuous inventory, stage-wise monitoring and regular audit.

Examination writing skeleton

1. Define fortification and explain its public-health purpose.
2. Identify Regulations 2 and 3, the heme-iron prohibition and the distinction between voluntary and mandatory fortification.
3. Present Schedule I salt, oil, milk, flour and rice standards with denominators and permitted sources.
4. Add optional nutrients and the multigrain-atta condition where space permits.
5. State the fortification declaration, prescribed logo, optional tagline and iodized-salt exception.
6. Quote the complete iron advisory.
7. List the twice-yearly undertaking and its quality assurance components.

8. Explain the negative tolerance and preserved maximum.
9. Conclude that label claims and the logo do not substitute for actual compliance.

8.22. FINAL LEGAL AND EXAMINATION CAVEATS

Issue	Final exam-safe position
Regulatory date	This handout audits the uploaded Version III compendium dated 30 September 2021; later amendments require separate verification
Fortification mandate	“When fortified” imposes conditions on fortified products; it does not by itself mandate fortification of every product in the category
August 2021 amendments	Distinguish their stated in-force date, 27 August 2021, from the stated compliance date, 1 March 2022
September 2021 label-tolerance amendment	Distinguish its stated in-force date, 22 September 2021, from the stated compliance date, 1 April 2022
Iodine	Use the substituted 15–30 parts-per-million dry-weight range when discussing the displayed August amendment; identify the earlier two-stage provision if asked historically
Iron source	Never transfer atta’s complete source list automatically to rice
Fortification tagline	Optional under Regulation 7(2), despite the mandatory declaration and prescribed logo requirements that otherwise apply
Iodized-salt exception	Limited to the requirements of Regulation 7(2) for iodized salt fortified with iodine; not a blanket exemption from food law
Logo geometry and Pantone specification	Not established by the supplied accessible Schedule II text; no dimensions or colour code are invented here
Iron advisory	Use the full substituted thalassemia and sickle-cell-anaemia statement
High Fat Sugar Salt exclusion	Expressly concerns the Fortified Processed Foods category
Schedule I tolerance	Minus 10% of declared total value, with the maximum added-micronutrient limit preserved; no assumed positive tolerance
Fortified milk powder	Government-programme restriction, reconstitution requirement and additional label statements apply under the displayed insertion
Frying-oil limits	The requested 15% and 25% targets require their governing instrument and exact boundary wording; the fortification compendium is not their authority
General colour limits	The requested 100- and 200-milligram-per-kilogram figures and exact product exceptions are not established by the supplied documents
Peroxide value	A primary-oxidation index, not proof of overall used-oil safety
Marking breakdowns	Suggested teaching rubrics, not official HPSC schemes

Issue	Final exam-safe position
Screening penalty	Calculations assume the user-supplied one-mark and minus-0.25 pattern; actual examination instructions prevail

Final revision rule: Attach every number to its food vehicle, chemical source, denominator, regulatory version and exception. Attach every analytical conclusion to the reaction that the assay actually measures.

End of the complete Section 8.18 continuation and requested remaining examination material.